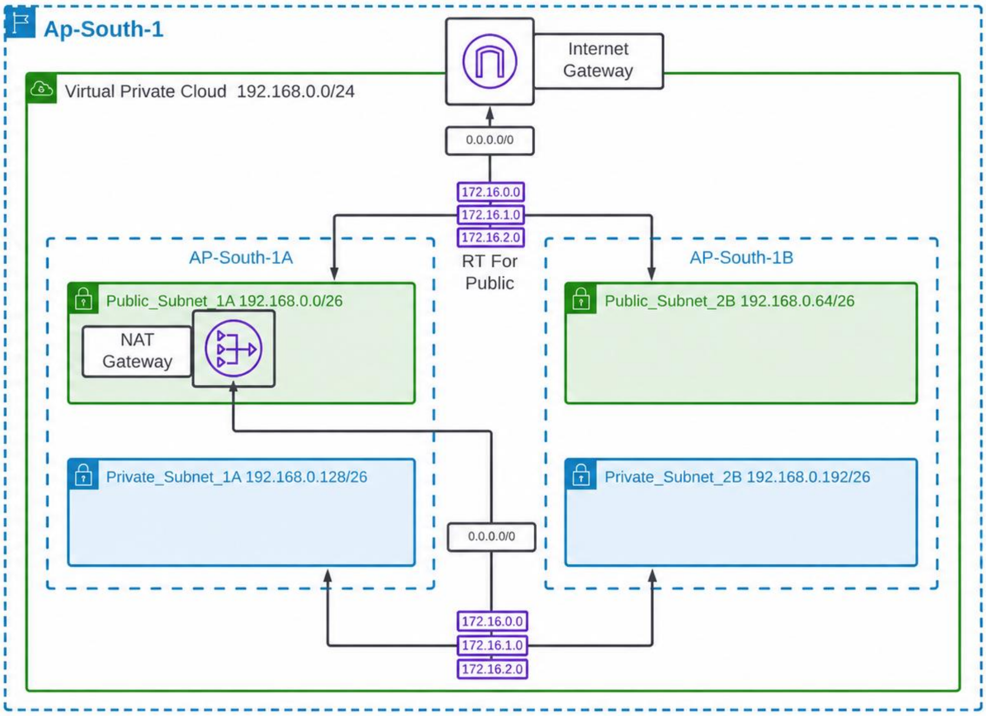
Network Load Balancer

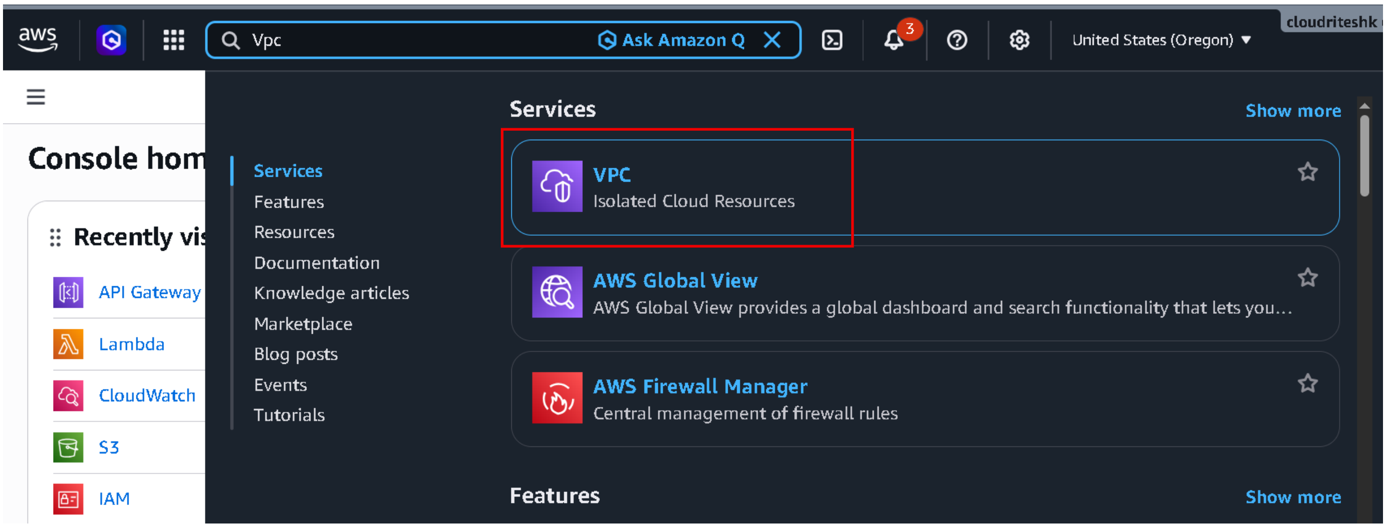
Summary: -

I worked on a Network Load Balancer where incoming traffic was distributed between multiple servers to improve speed, handle high traffic, and keep the application running smoothly. I also configured networking, security, and server connectivity for better performance and availability.

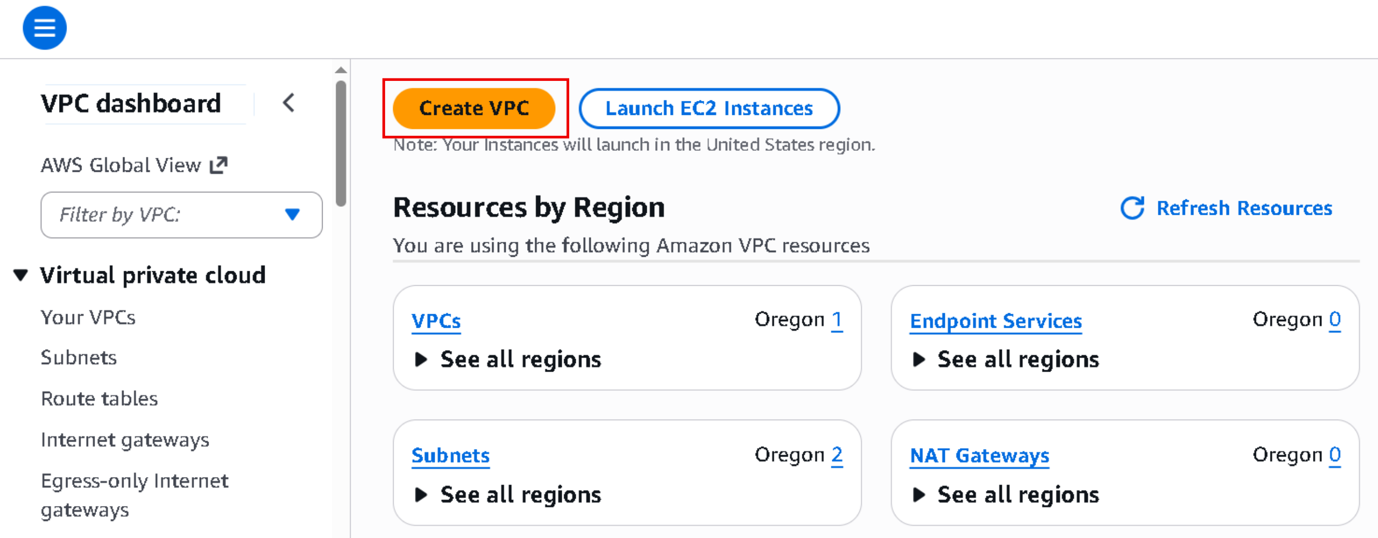
Implementation Load Balancer VPC Design: -



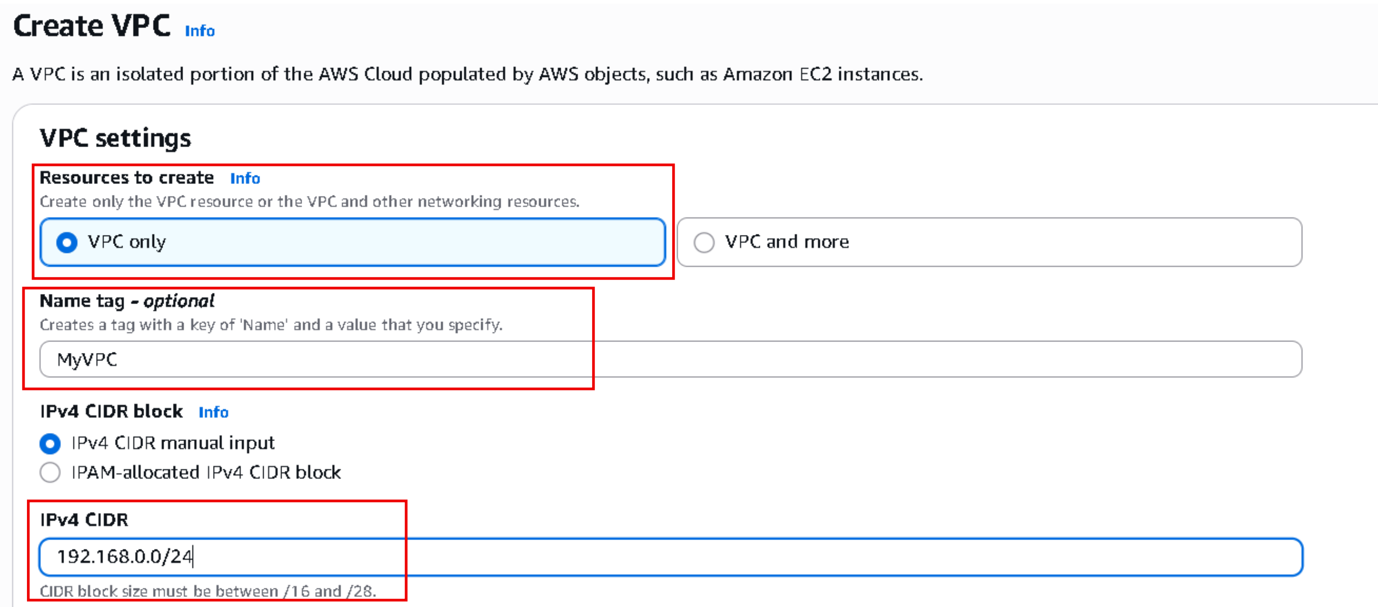
- Firstly, we create our VPC.
- Go to your AWS Console.



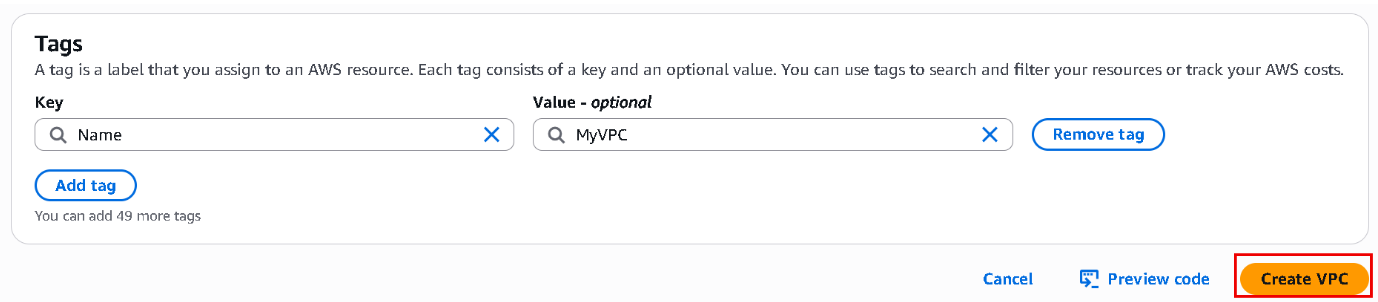
- Now Search VPC.
- And open it VPC.



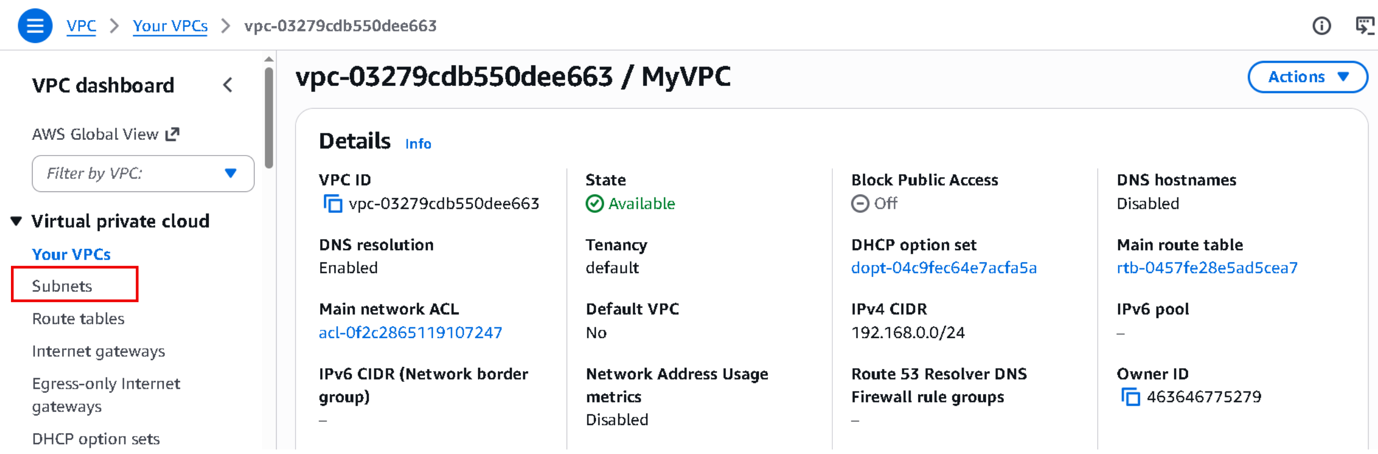
- Now simply click Create VPC.



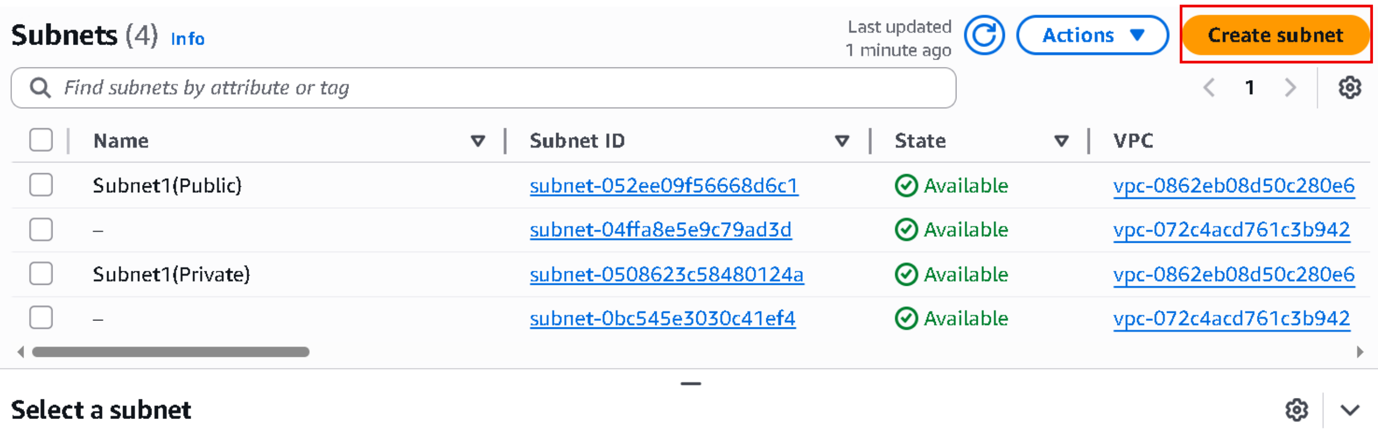
- Select VPC only.
- Then Enter your VPC name.
- And then Enter IPv4 CIDR as 192.168.0.0/24



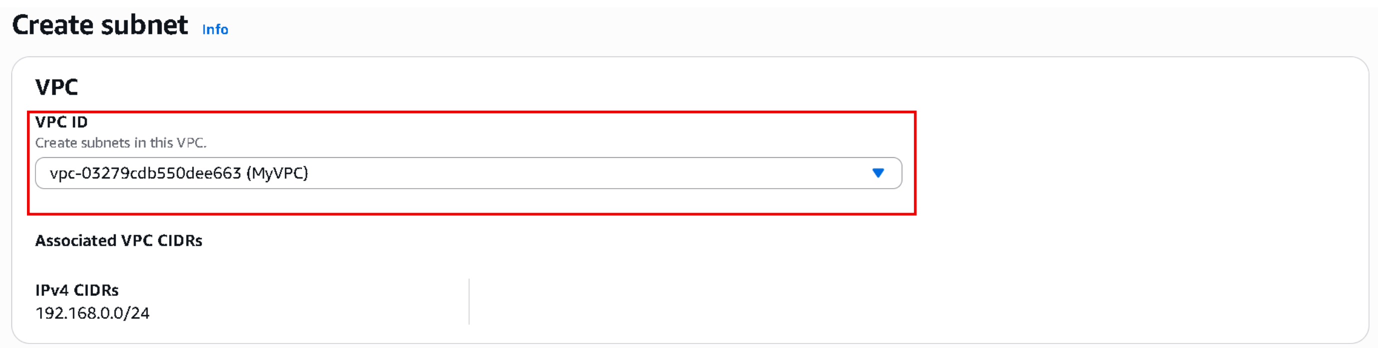
- Now simply scroll down and click Create VPC.
- Now we need to create four Subnet.
- We are creating two public subnets (public subnets having inbound internet access allowed).
- And we create two private subnets (which have no inbound internet access allowed).



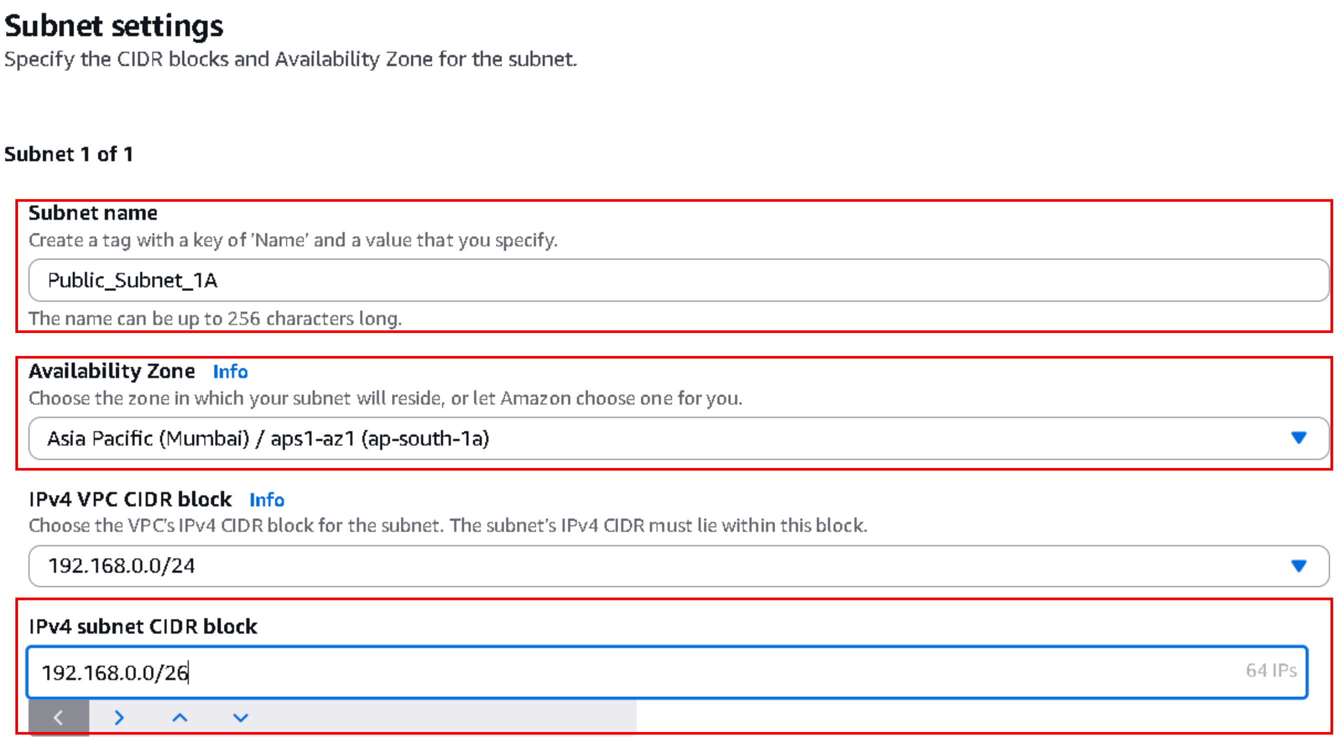
- Now click on subnets.



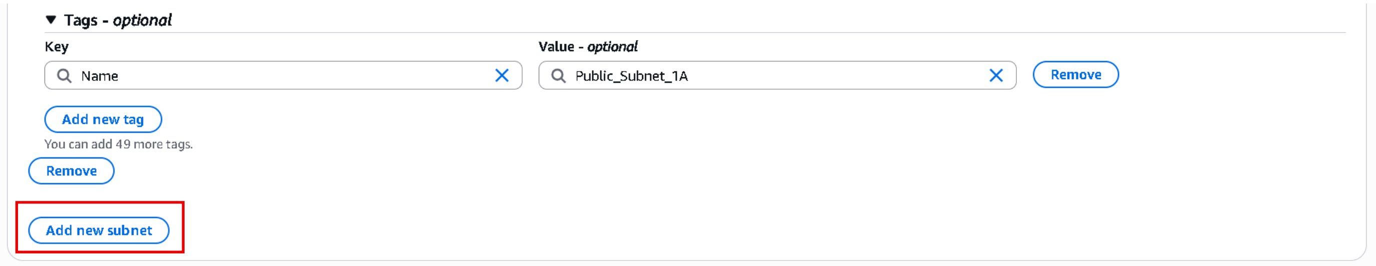
- Now simply click on Create subnet.
- Firstly, we create two public subnets.



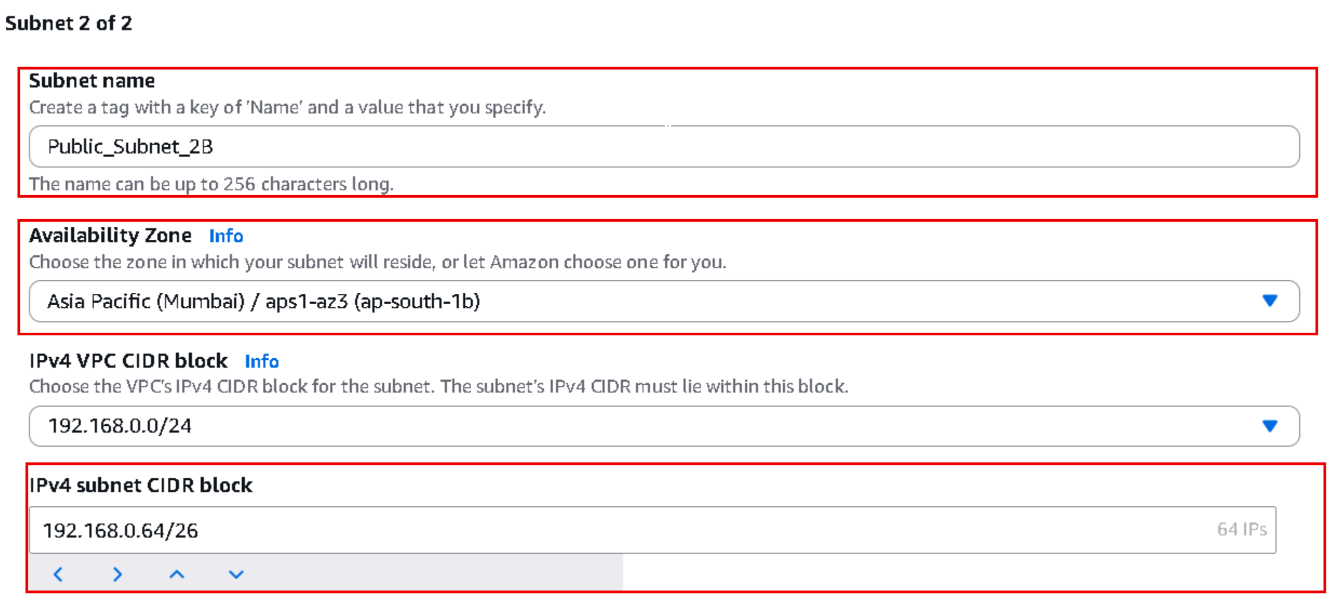
- Select your VPC that you created now.



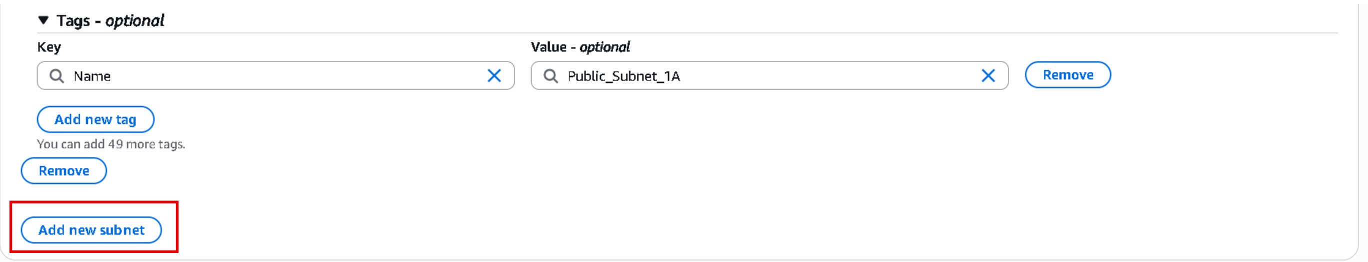
- Enter Subnet name as Public _Subnet_1A
- The select Availability Zone (ap-south-1a).
- Then IPv4 subnet CIDR block as 192.168.0.0/26
- Now simply scroll down.



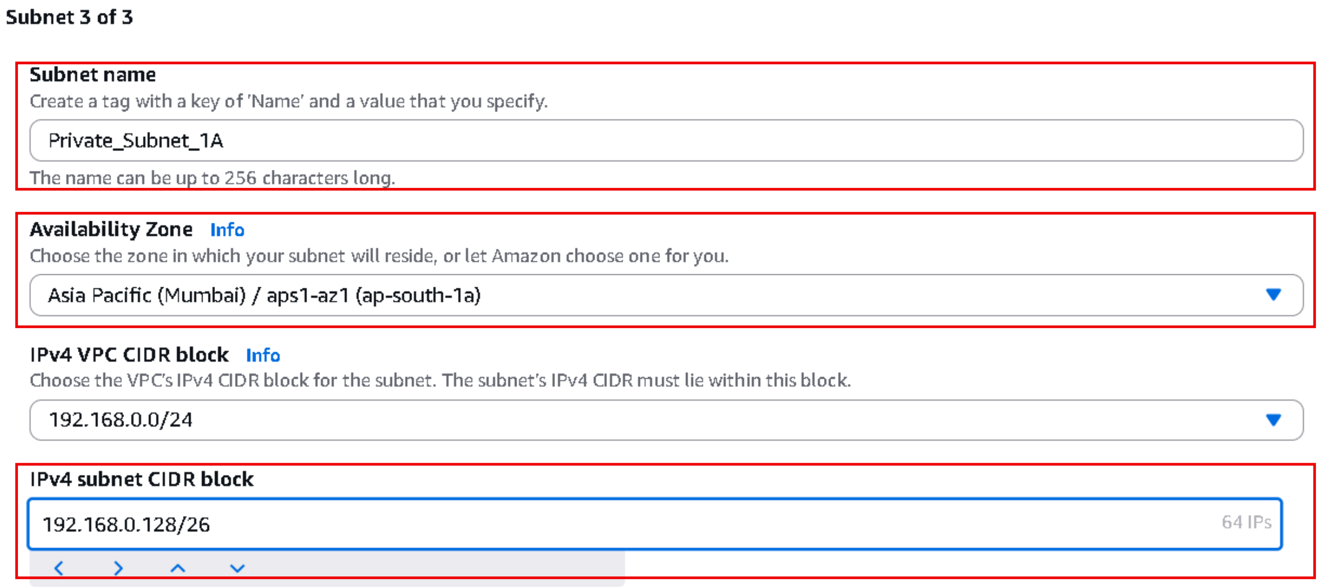
- Now we are going to create another click Add new Subnet.



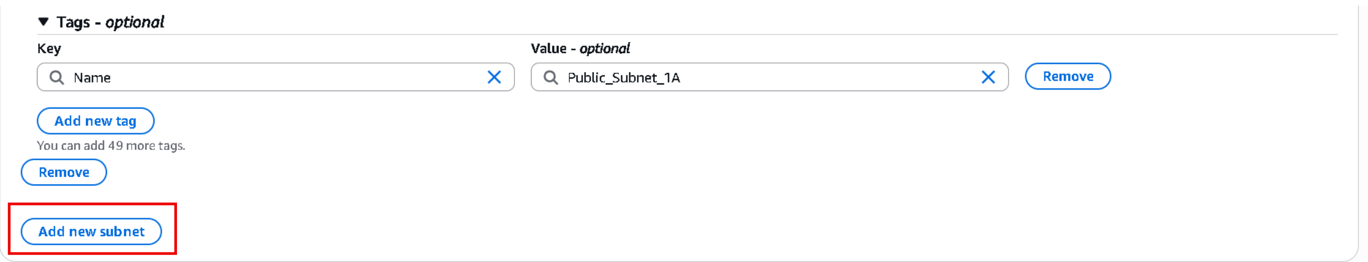
- Enter subnet 2 name as Public_Subnet_2B
- Select Availability Zone (ap-south-1b)
- Then Enter IPv4 subnet CIDR block as 192.168.0.64/26
- Now simply scroll down.



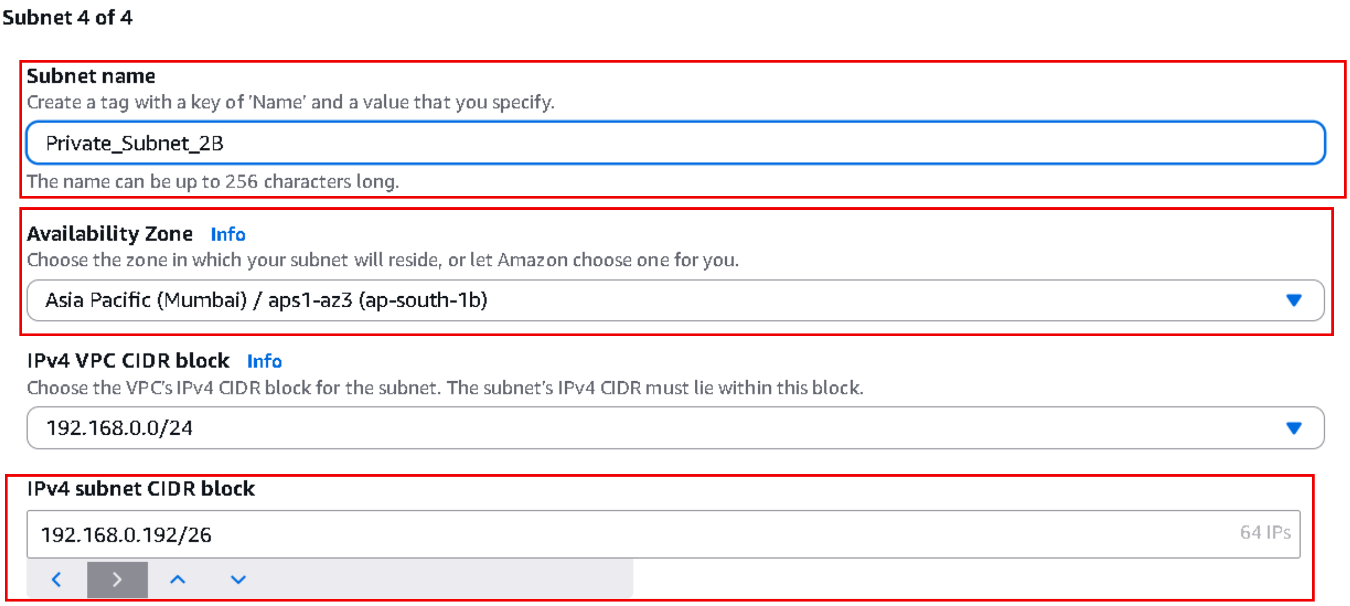
- Now click Add new subnet.
- Now we will add Private Subnets.



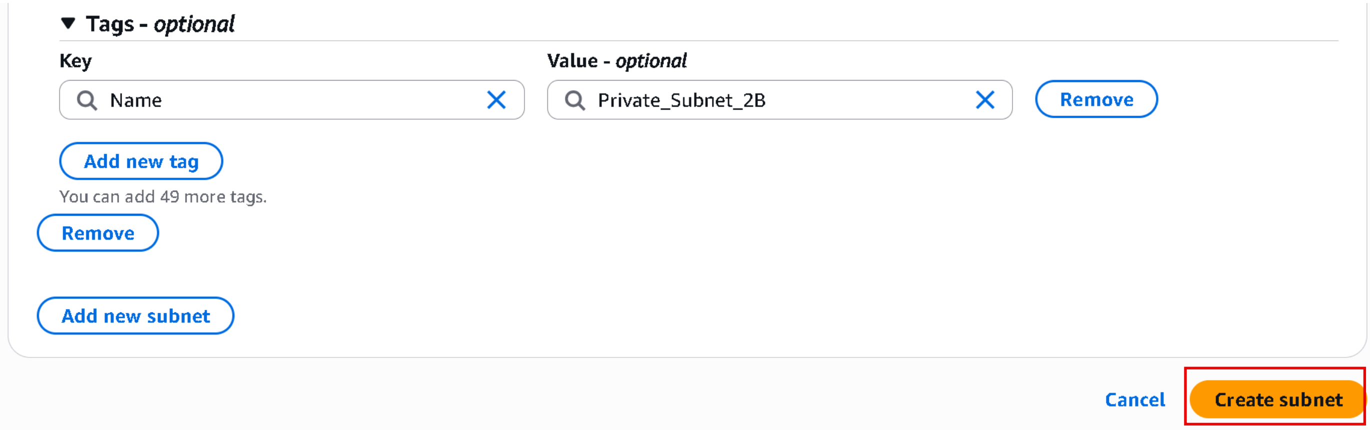
- Enter Subnet name as Private _Subnet_1A
- The select Availability Zone (ap-south-1a).
- Then IPv4 subnet CIDR block as 192.168.0.128/26
- Now simply scroll down.



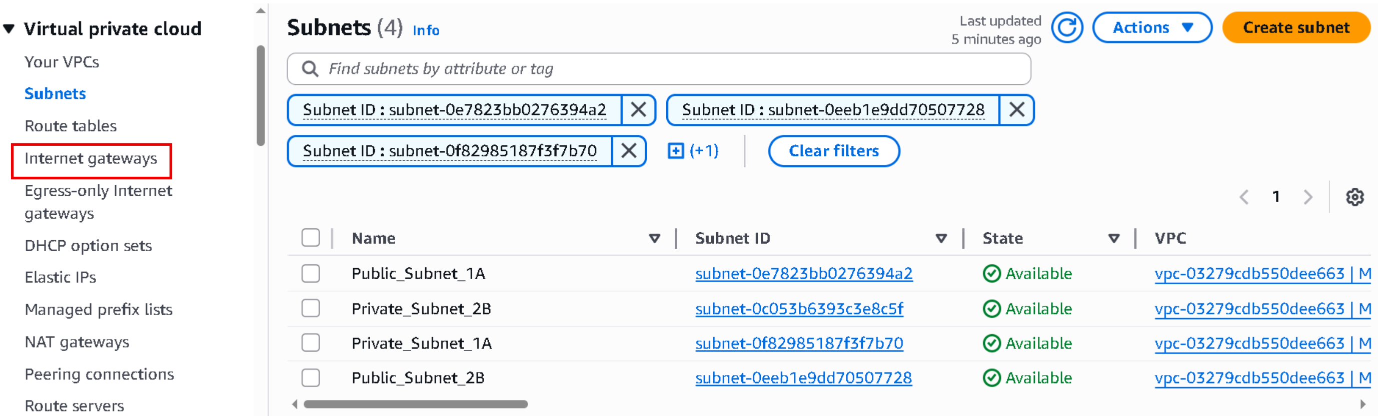
- Now we are going to create another click Add new Subnet.



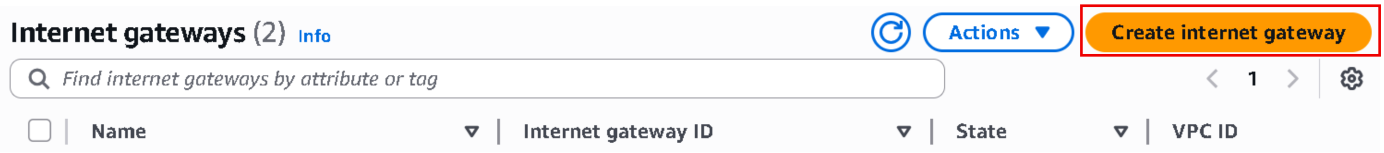
- Enter Subnet name as Private _Subnet_2B
- The select Availability Zone (ap-south-1b).
- Then IPv4 subnet CIDR block as 192.168.0.192/26
- Now simply scroll down.



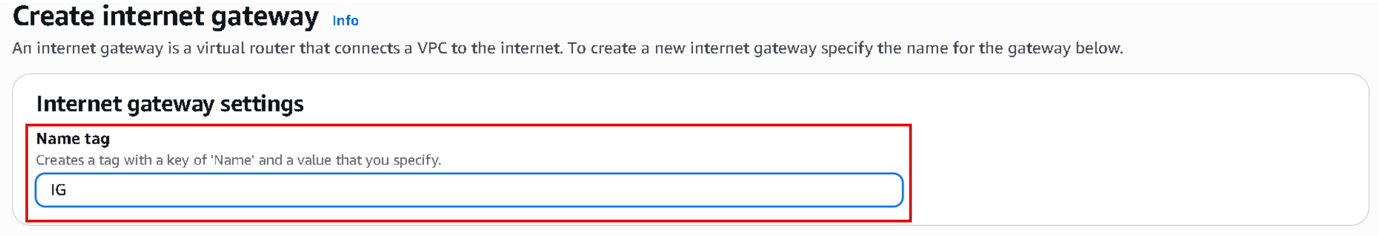
- Now click Create subnet.



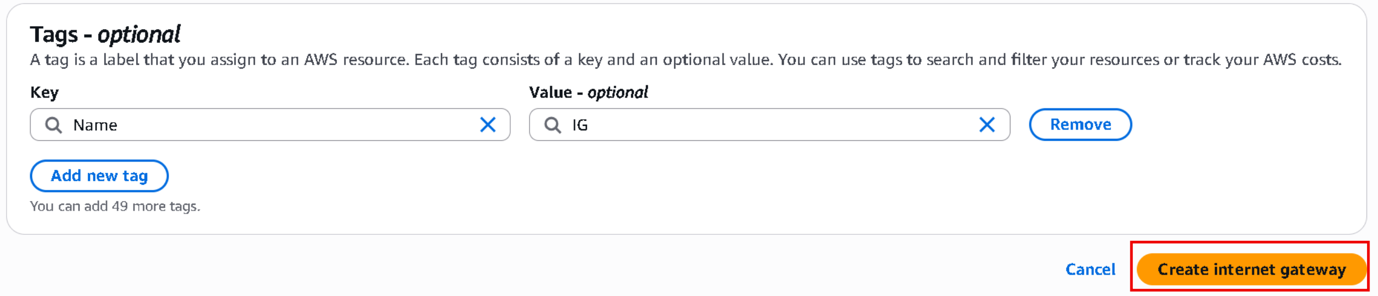
- Now our Four Subnet is created.
- Now to provide Internet connectivity, we have to add Internet gateway and we have to attach internet gateway with our VPC.
- Now click Internet gateways.



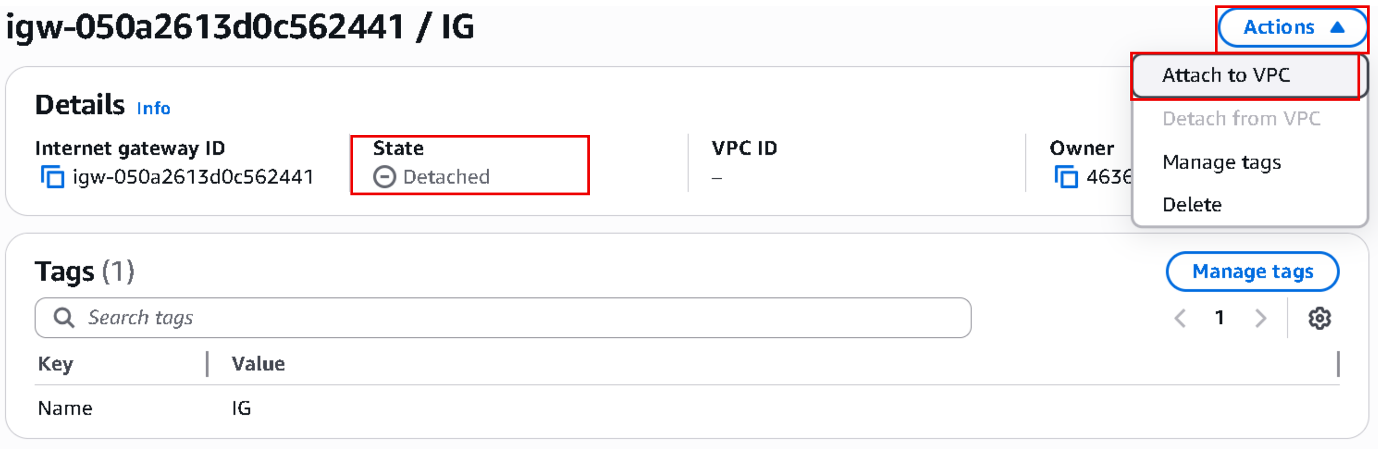
- Now click Create internet gateway.



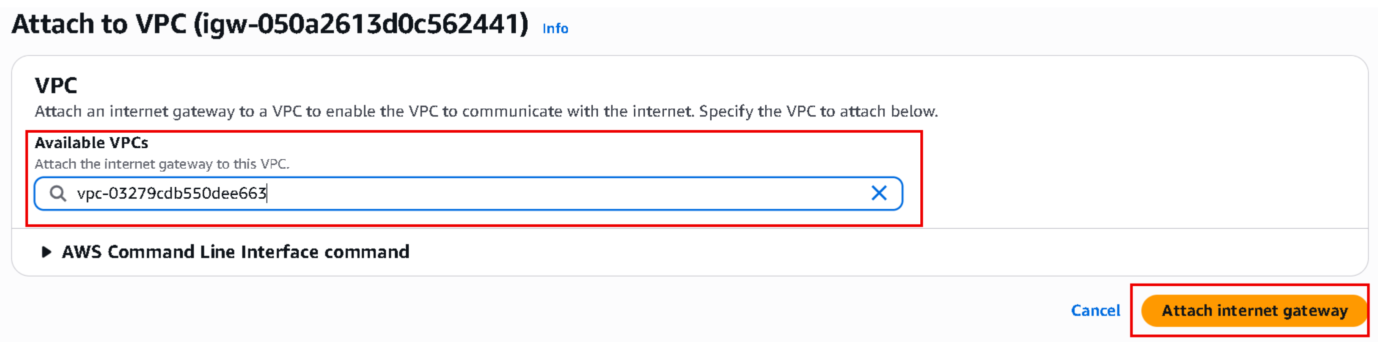
- Now Enter Name (IG).



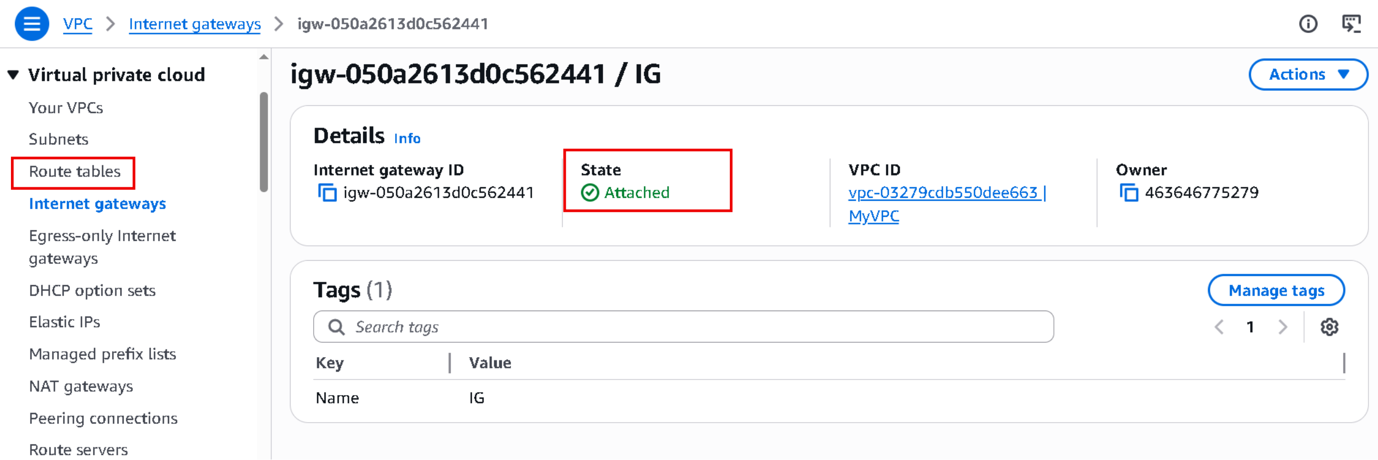
- Now simply click Create internet gateway.



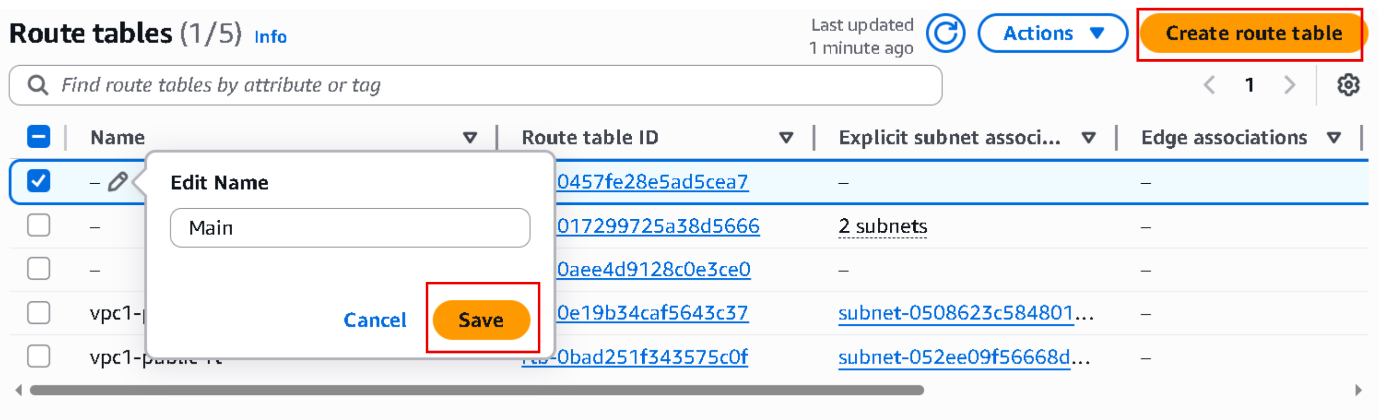
- Now our internet gateway is created.
- But you see it's not Attach with our VPC.
- Now click Action and then select Attach to VPC.



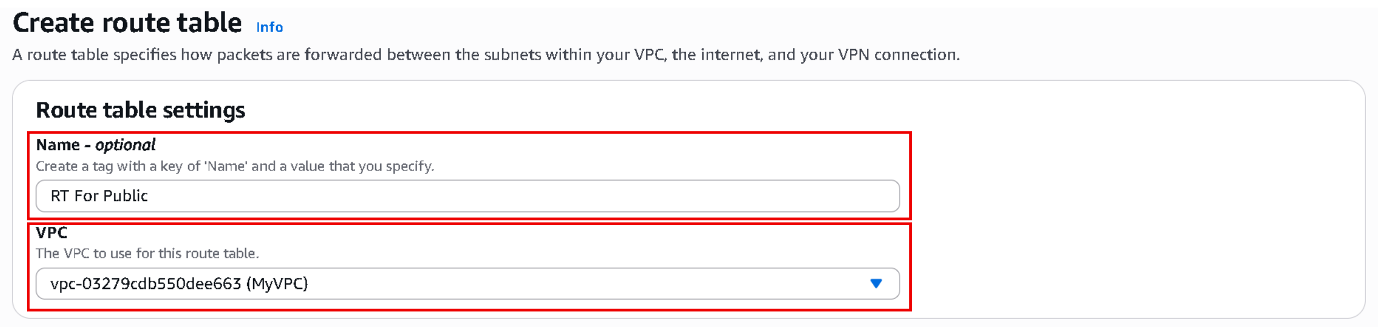
- Now Select Create VPC and then Click Attach internet gateway.



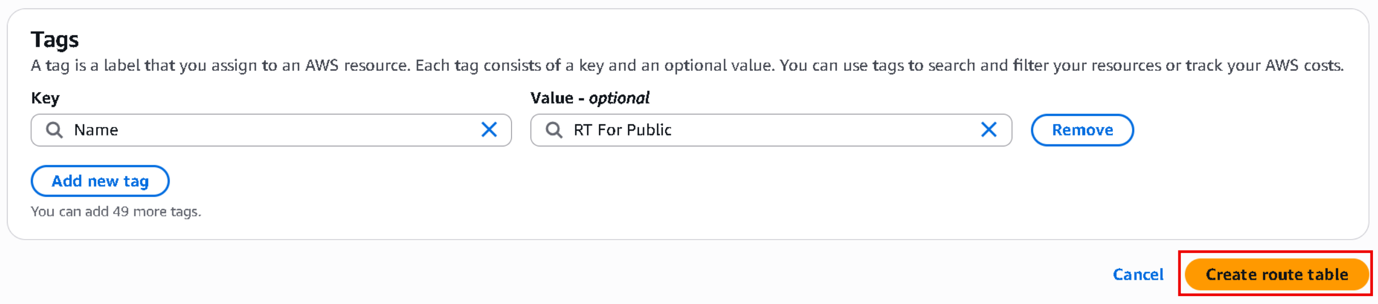
- it’s Attach to VPC.
- Now we have to set up route table.
- Now we have a main route table which is default and we want to provide inbound connectivity to this public subnet.
- So, for this I am going to create Separate Route table for public subnet
- Now click on Route tables in nav bar.



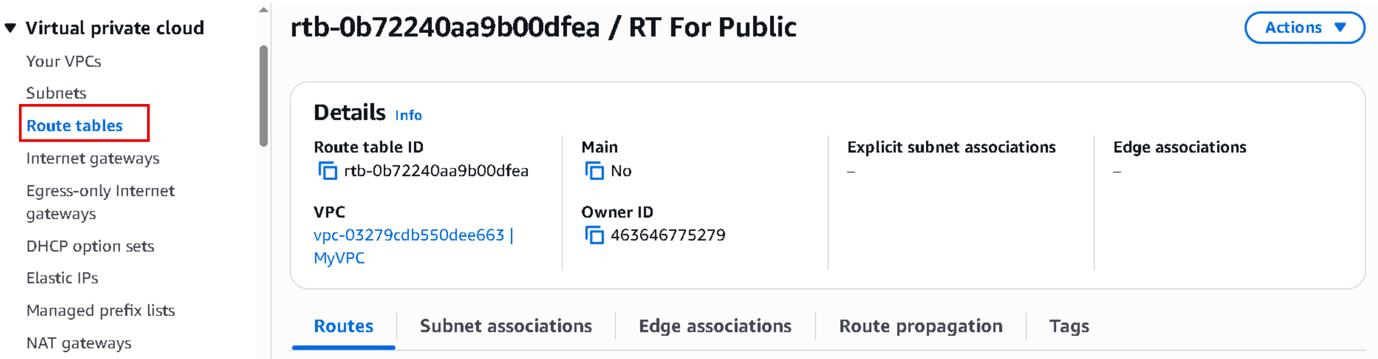
- Now 1st one is our Default Route table we name it as Main and save it.
- we know that this is our VPC default Route table by clicking Route table ID.
- Now click on Create route table.



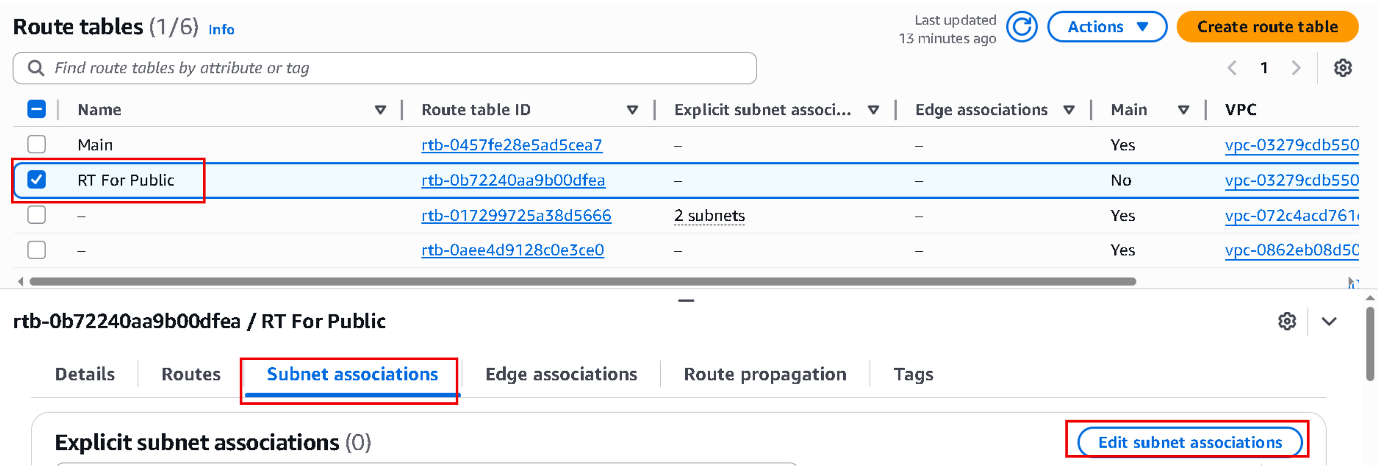
- Enter name of Route table (RT For Public)
- Now select VPC that you created now.



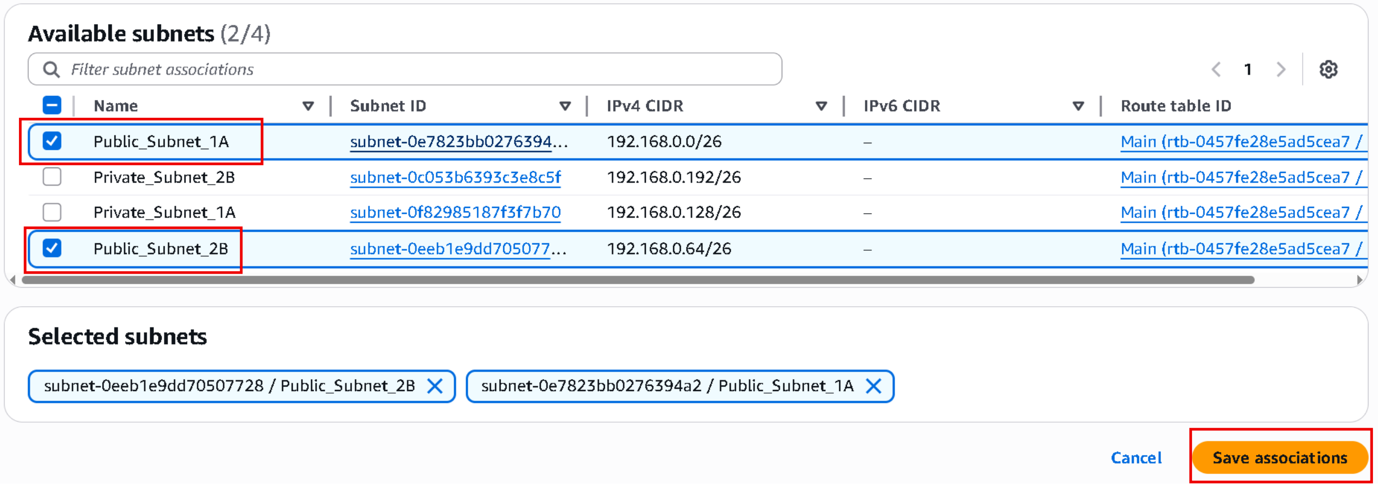
- Now simply scroll down and click Create route table.



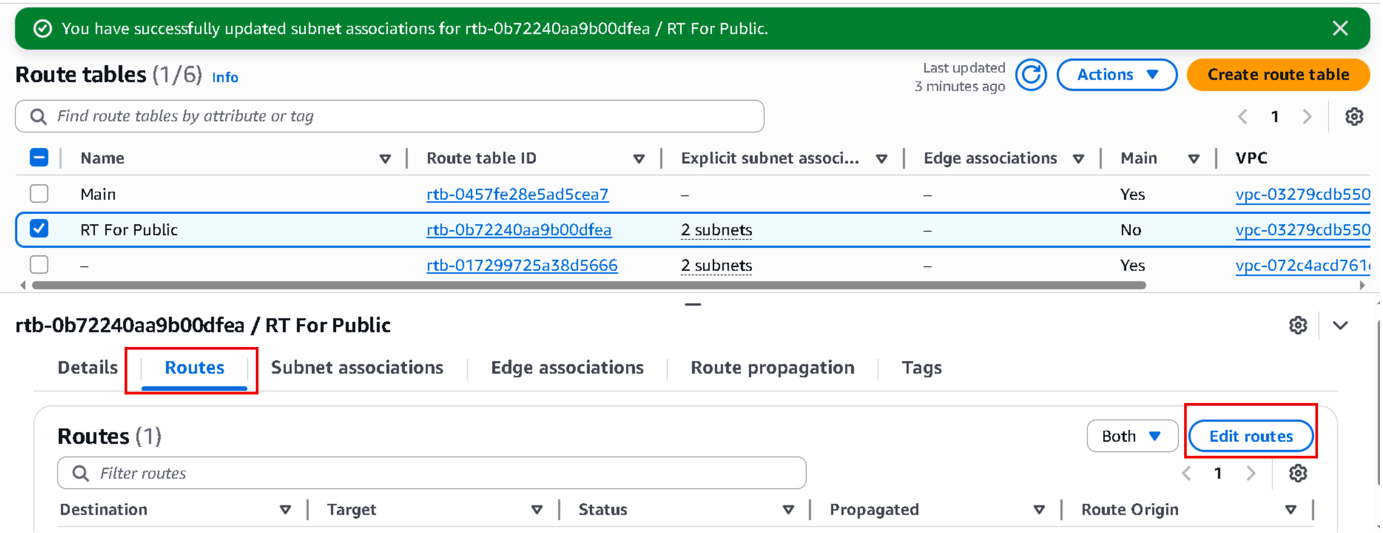
- Now our route table is created.
- But right now, your route table is not associated with public subnets.
- Now click on Route table.



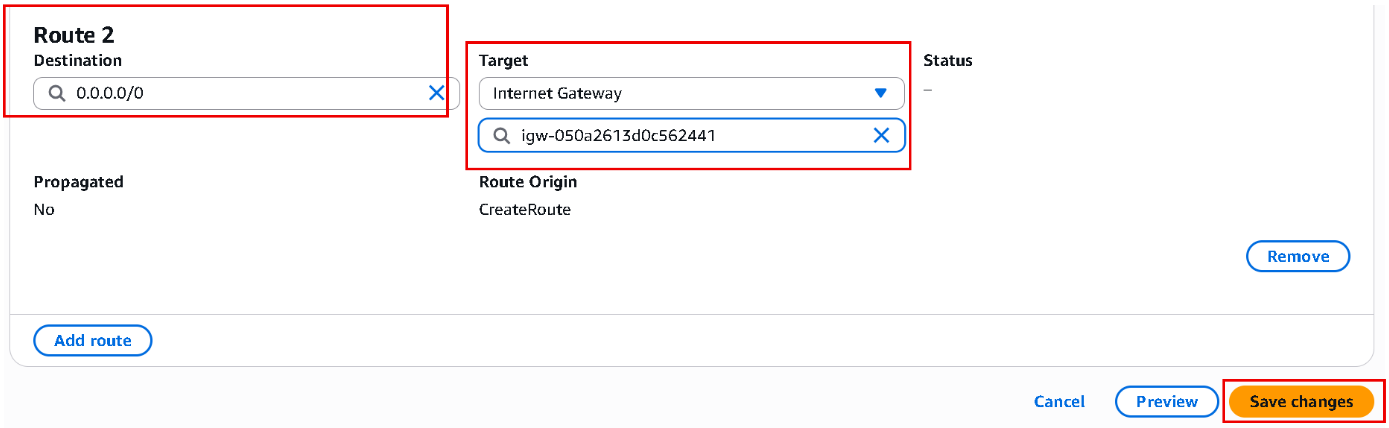
- Now select RT For Public route table.
- And click Subnet associations.
- And then click Edit subnet association.



- Now select both public Subnets.
- And click Save association.
- Now we have to add this particular route towards the internet gateway into the route public.

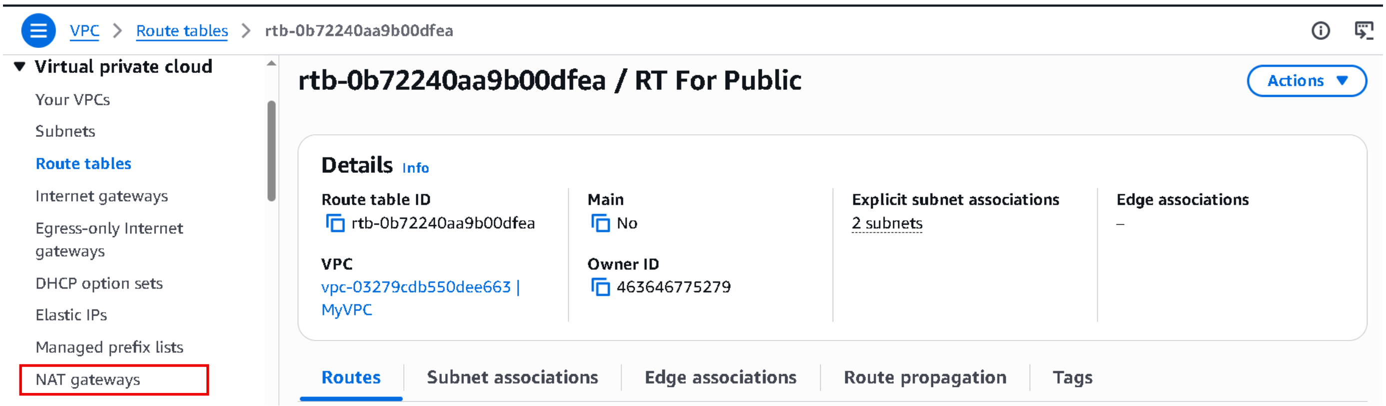


- Now click Routes and then click Edit routes.

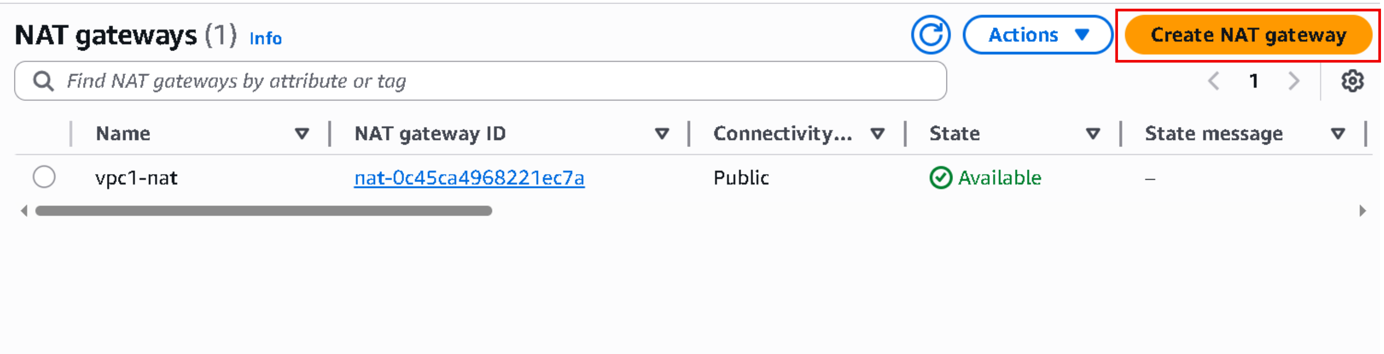


- Now Firstly, click Add route and then Enter 2nd Route details.
- Select Destination 0.0.0.0/0
- Select Target Internet Gateway
- Then select Created Internet Gateway
- And simply click Save Changes.
- Now this is done and we need to Setup NAT Gateway.

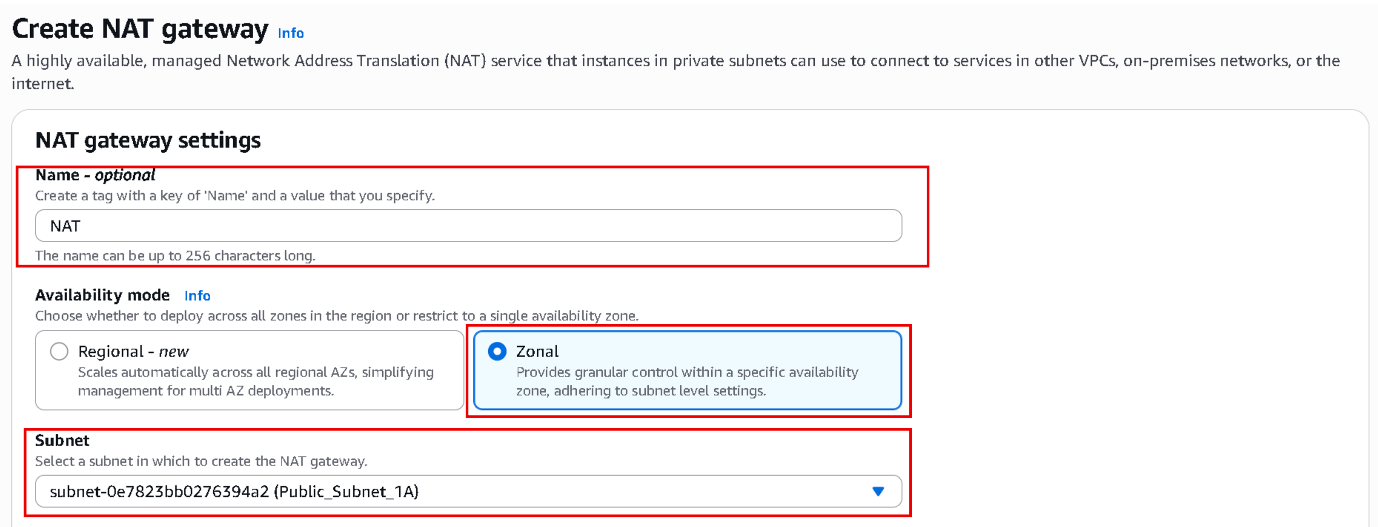
- Because you will place your Ec2 instance into the private subnet.
- If we are not going to set up NAT Gateway then you cannot provide outbound internet access to your EC2 instance.
- It means you cannot update your Ec2 instance.
- So, we have to create this NAT Gateway inside the public subnet.



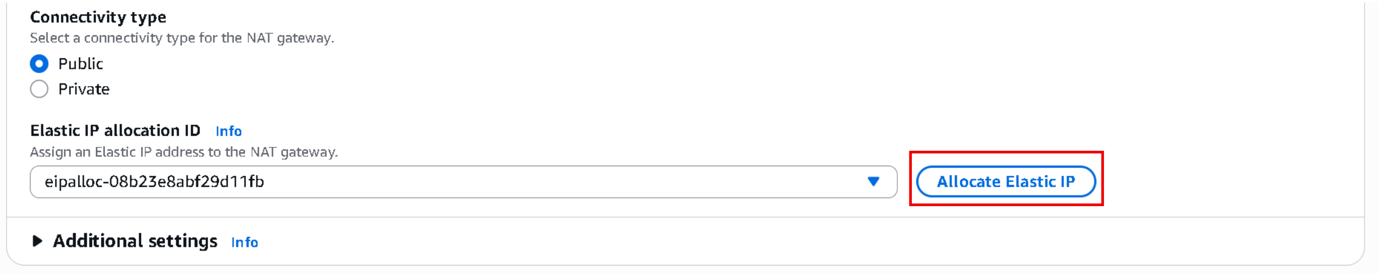
- Now click NAT Gateway in nav bar.



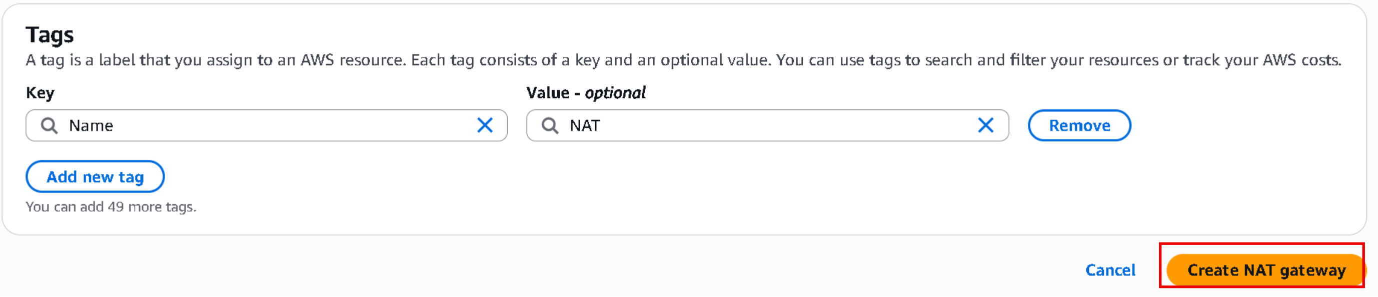
- Now click Create NAT gateway.



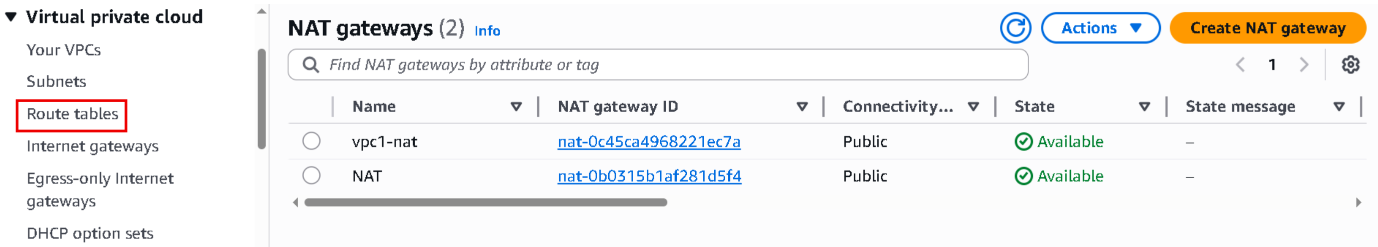
- Now Enter Gateway name.
- Availability mode as Zonal.
- Select Public_Subnet_1A
- Now scroll down.



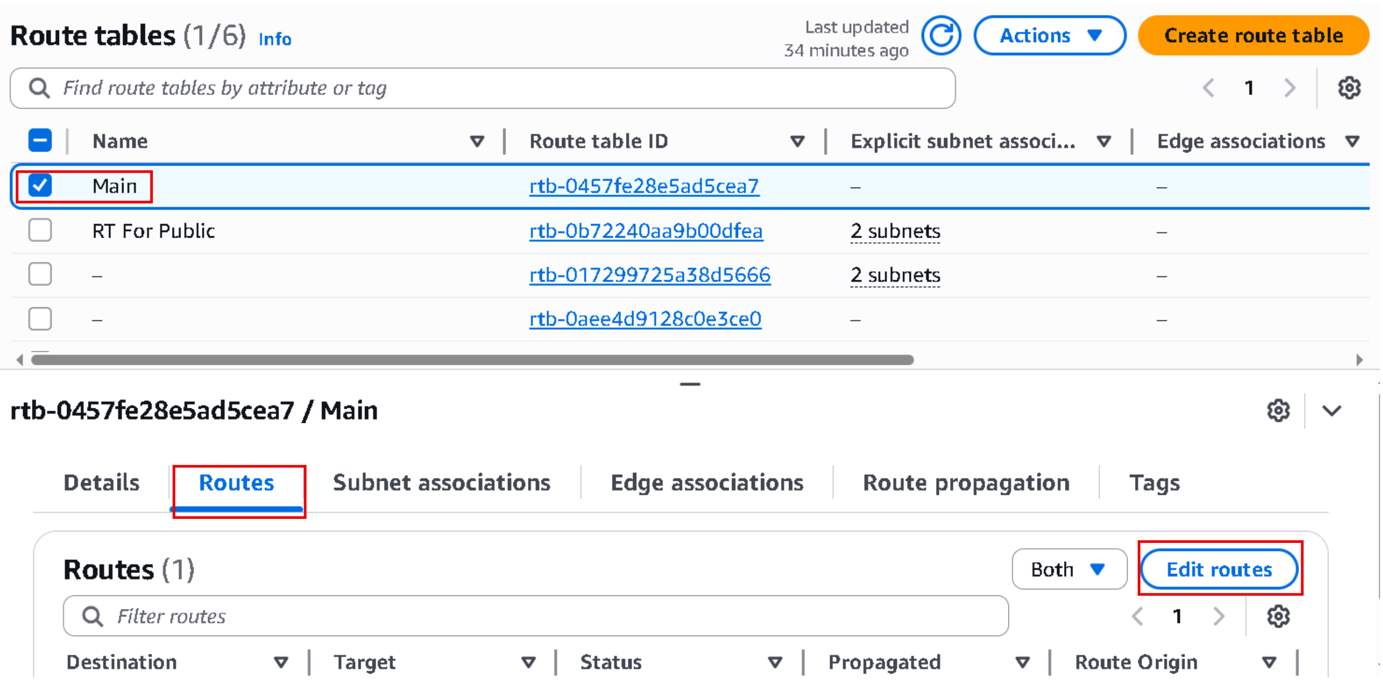
- Now click Allocate Elastic IP.
- Now scroll down.



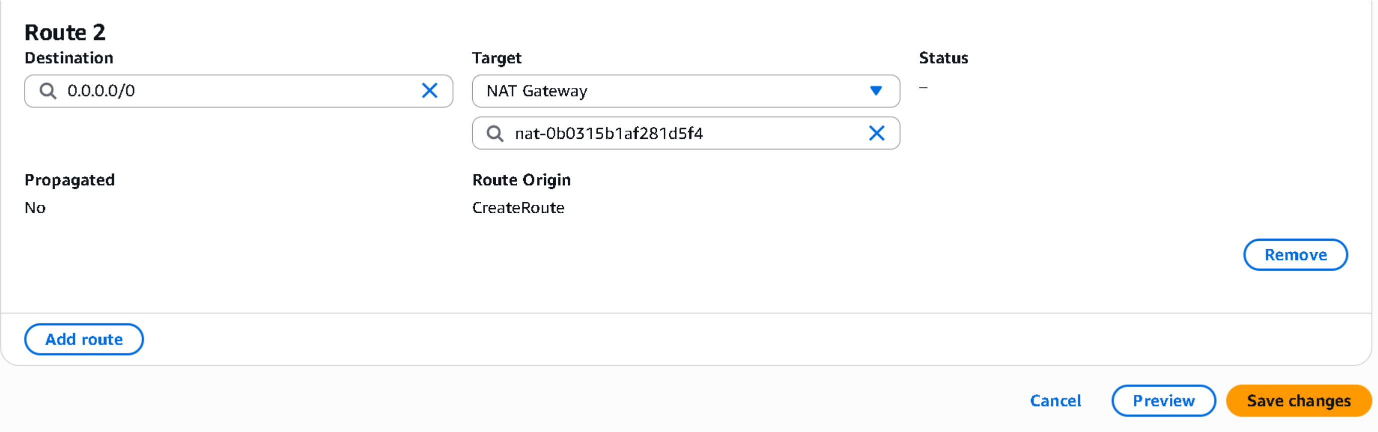
- Now click Create NAT gateway.
- Now we have to setup this 0.0 towards the net gateway.



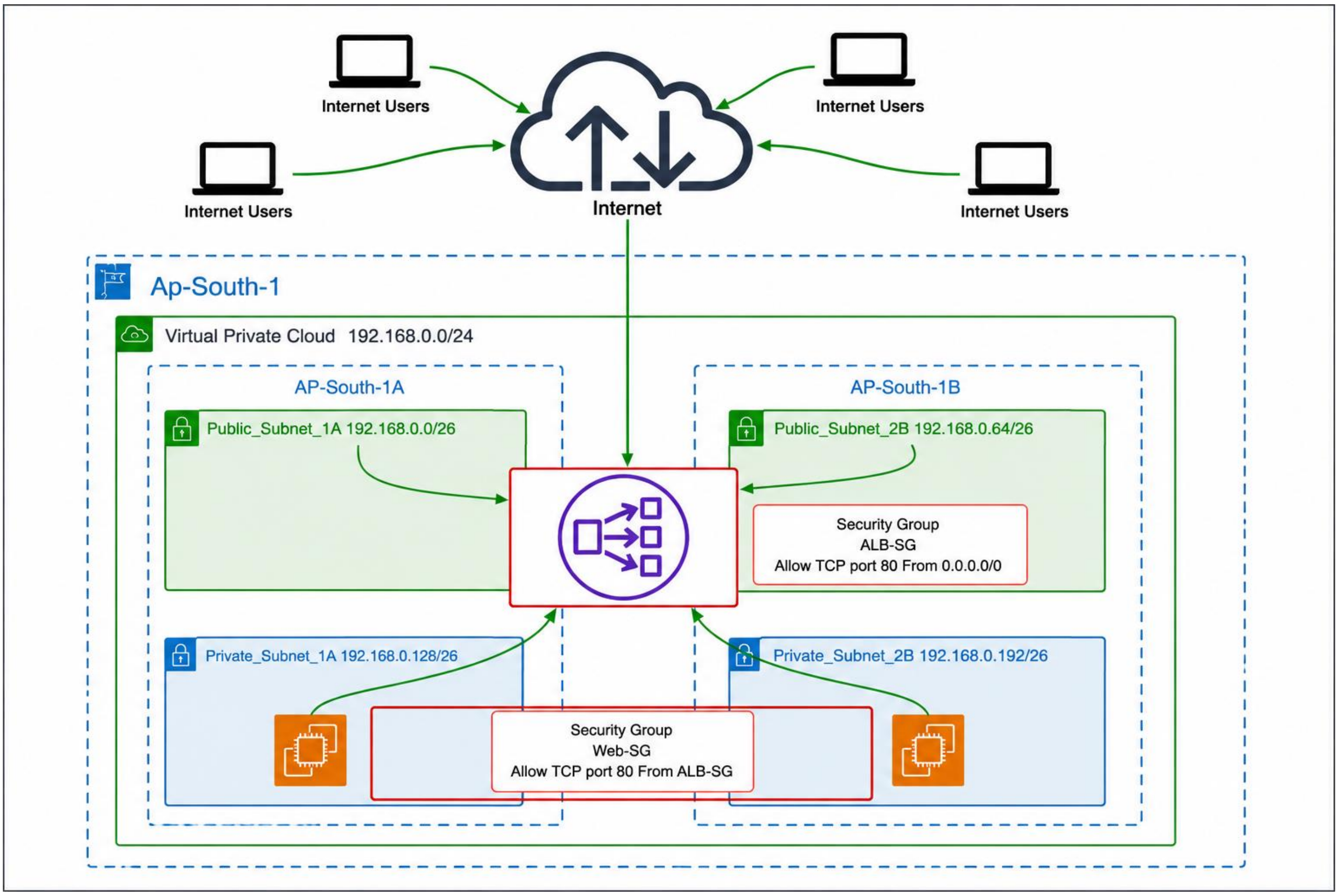
- Our NAT gateway is created.
- Now we have need to connect the NAT Gateway to our Main Route table.
- For we go inside route table for that click Route tables.



- Now select Main.
- And click Routes and then click Edit routes.

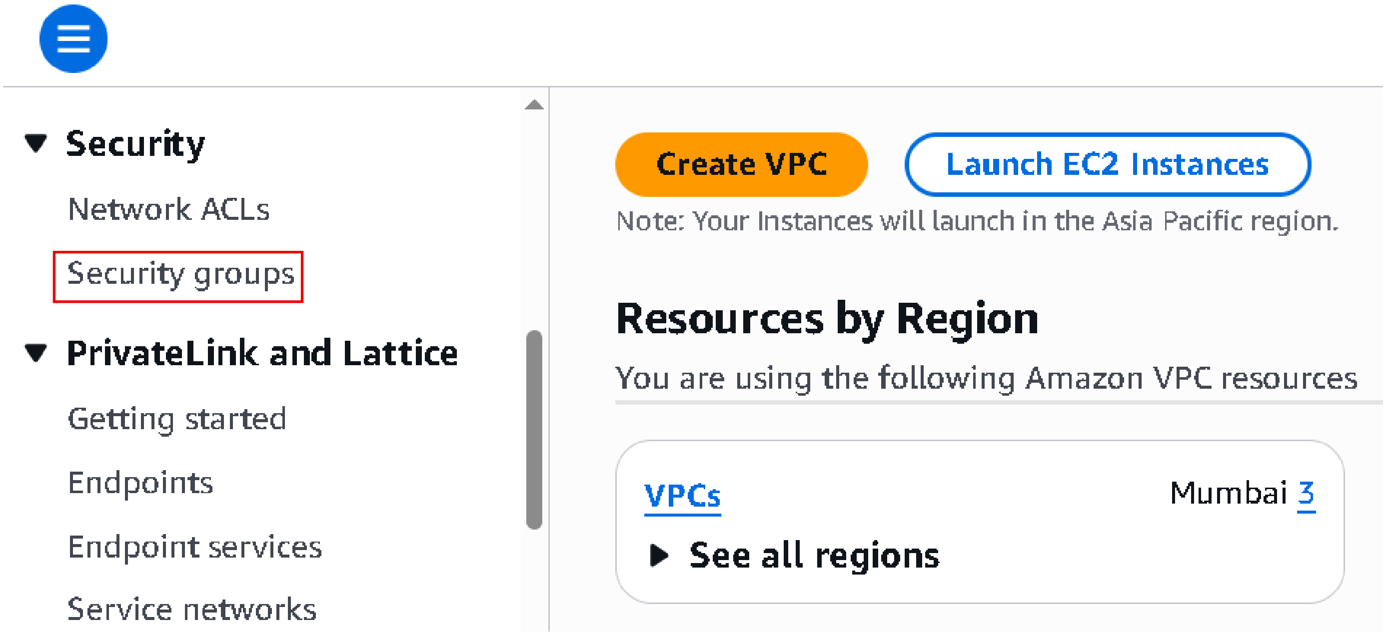


- Now Firstly, click Add route and then Enter 2nd Route details.
- Select Destination 0.0.0.0/0
- Select Target NAT Gateway
- Then select Created NAT Gateway
- And simply click Save Changes.
- And now we are ready to implement load balancer.
- As per rule when you are going to setup your load balancer you have to follow VPC best practices.
- And whatever we are doing right now we are setting up this according to the AWS best practices.

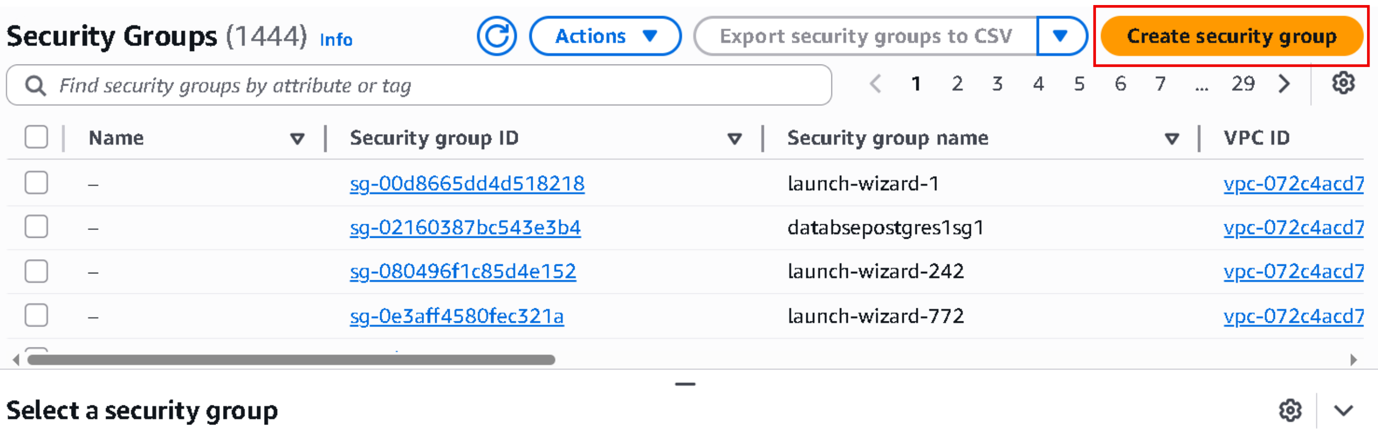


- So, in this image you see users from all over the internet will send requests to my load balancer URL.
- And then my load balancer will send this request to the EC2 instance.

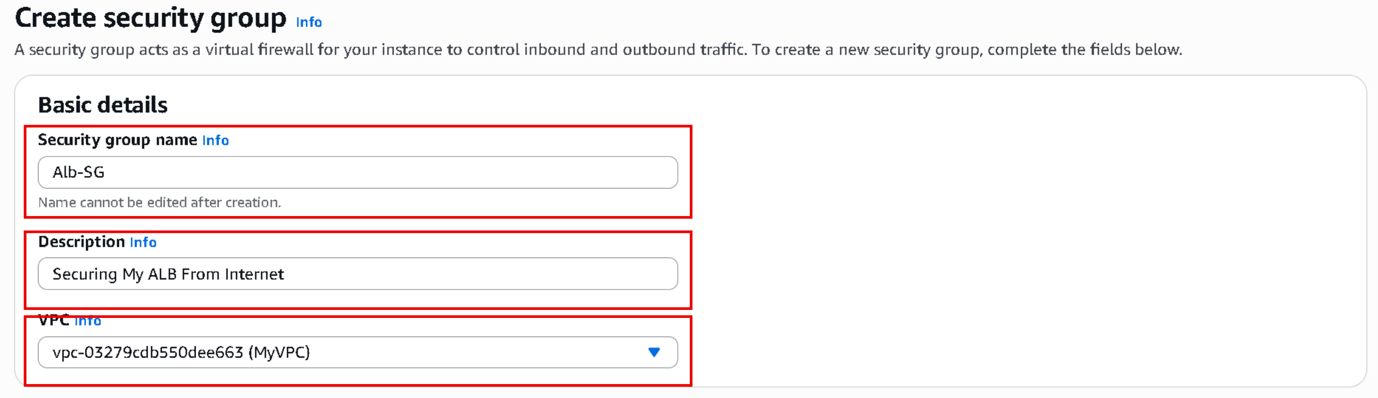
- We have to setup security group at two level.
- The first level we have is to set up a security group for load balancers(Alb-SG).
- And as we know that users over the internet will send requests to your load balancers on a port number 80.
- So, I just created other security group web-SG allows TCP port no 80 from(Alb-SG) anywhere because anybody over the internet can open website.
- And Second level we set up security group for the source(web-SG).
- Because accept the request if the request is coming from only load balancer.
- First we create our Security group then two EC2 instance.
- Then both thing done I create our Network Load balancer.



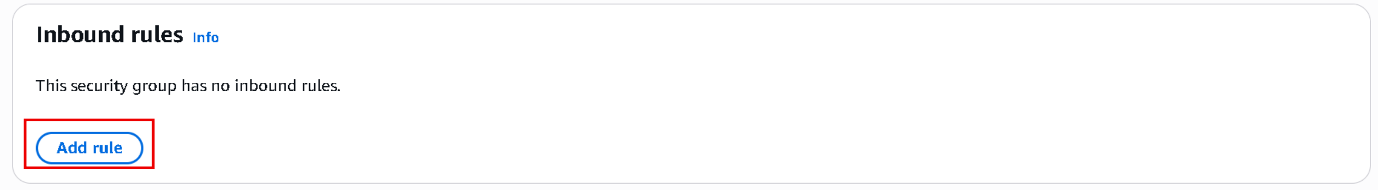
- Now opens VPC and in Nav bar scroll down.
- And click Security groups in Security.



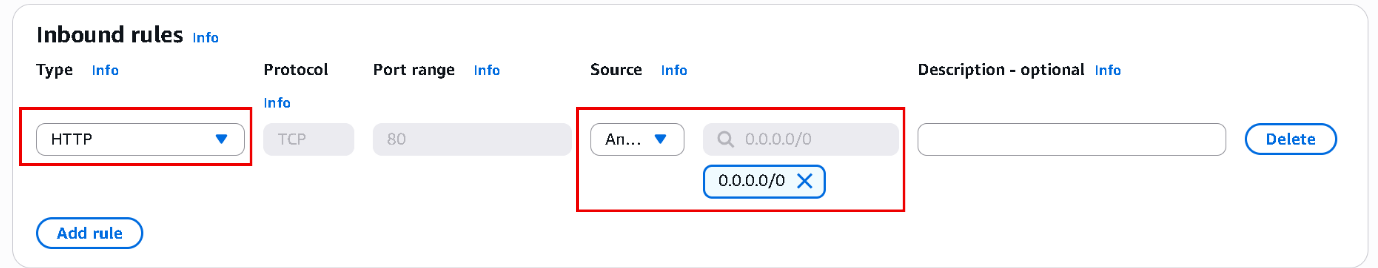
- Click security groups.



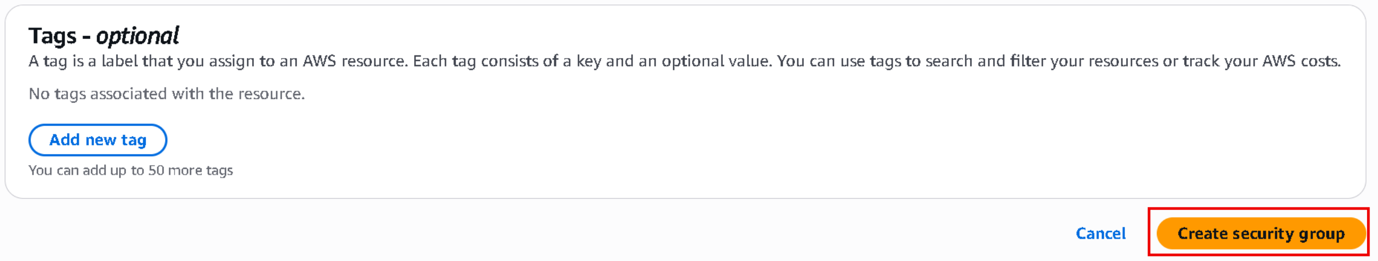
- Enter security group name (Alb-SG).
- Then Enter description (Securing My ALB From Internet).
- And then select Create VPC.



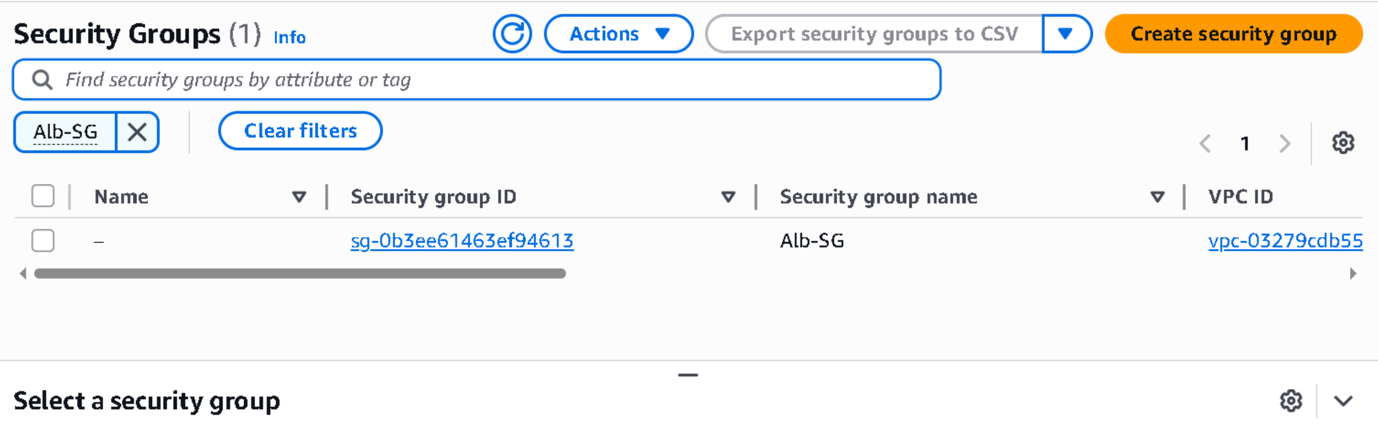
- Now click Add rule.



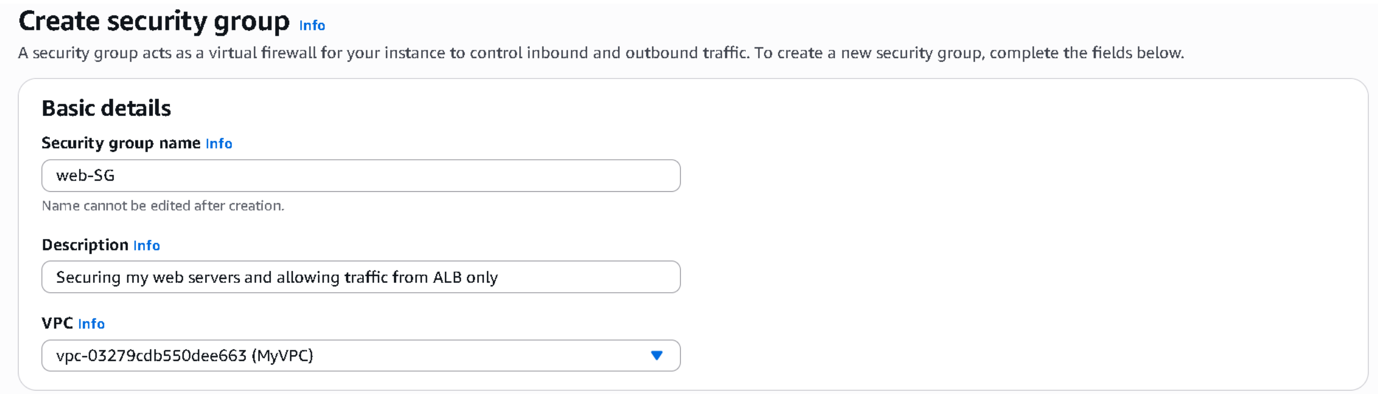
- Now select type as HTTP.
- And Select source From Anywhere.
- Why anywhere because we are going to get request from internet so that anybody can open my website.



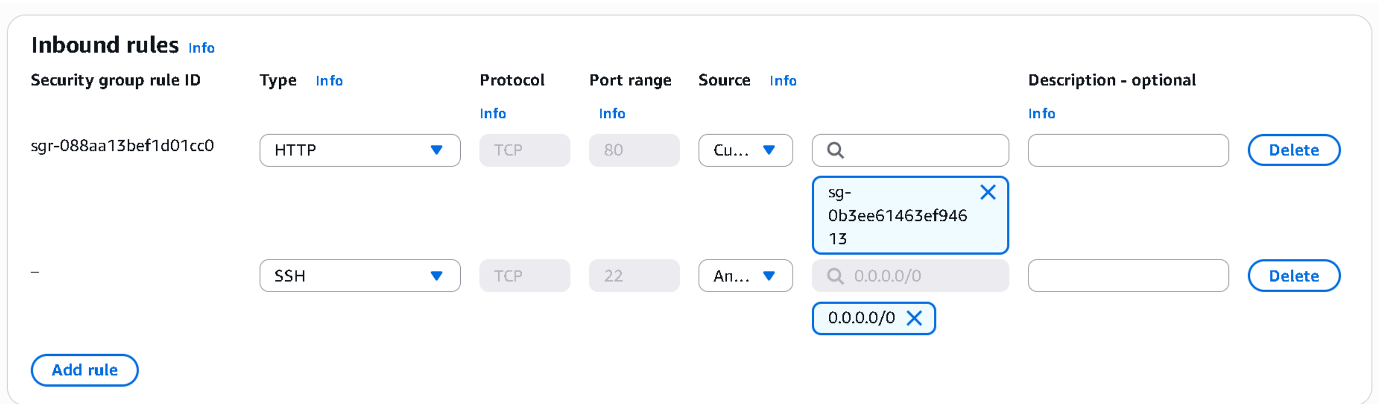
- Now simply scroll down and click Create Security group.



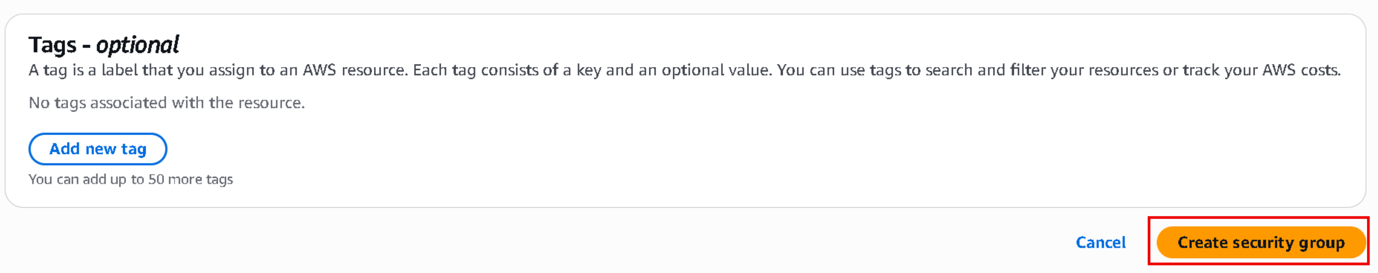
- First, Security group is created.
- Now we create 2nd Security group.
- So, click create security group.



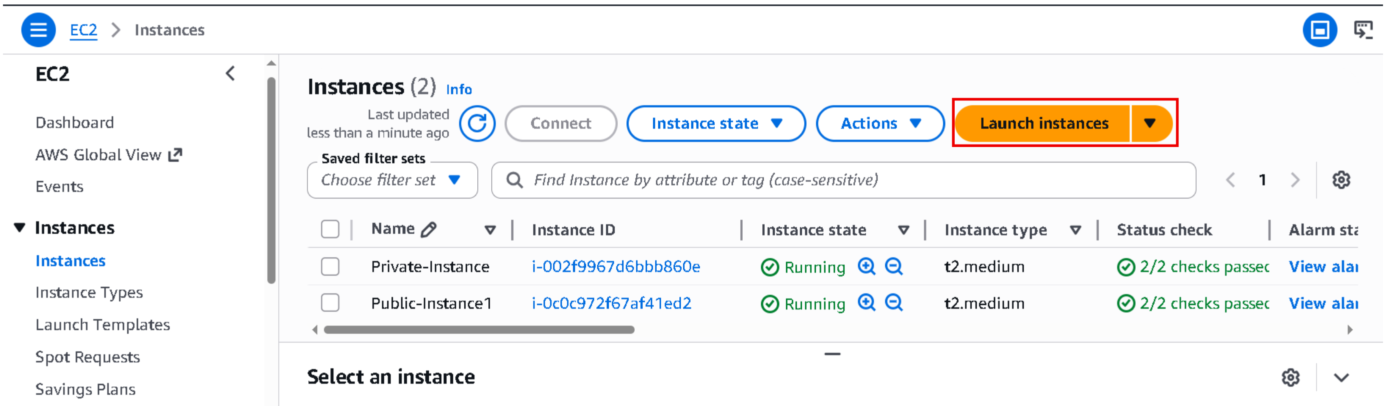
- Enter Security group name (web-SG)
- Enter Description(Securing my web servers and allowing traffic from ALB only).
- Select your Created VPC.



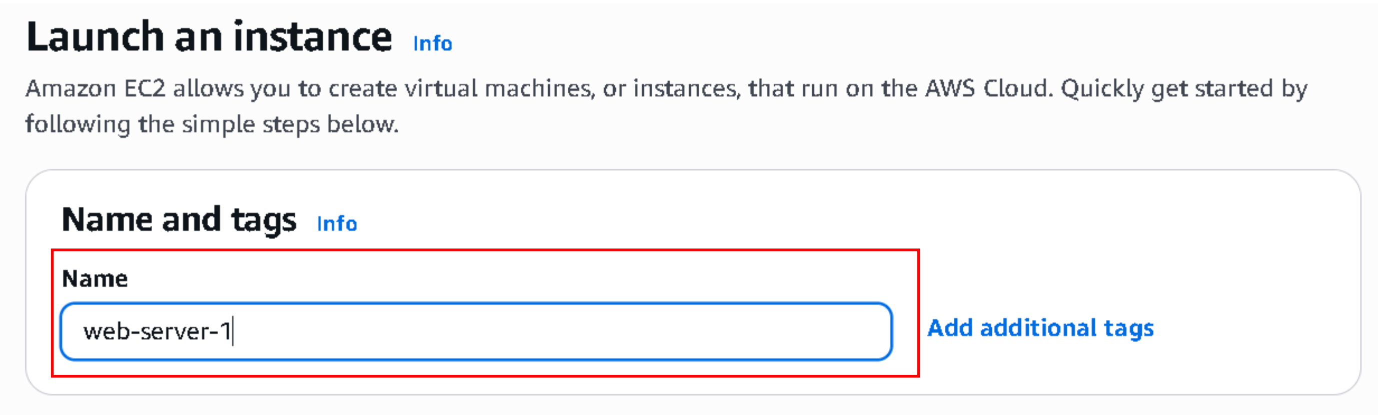
- Now add Inbound rules.
- Select type HTTP.
- Then Select Source Alb-SG security group that you created.
- And then add one more inbound rules.
- Select type SSH
- And then source Anywhere.



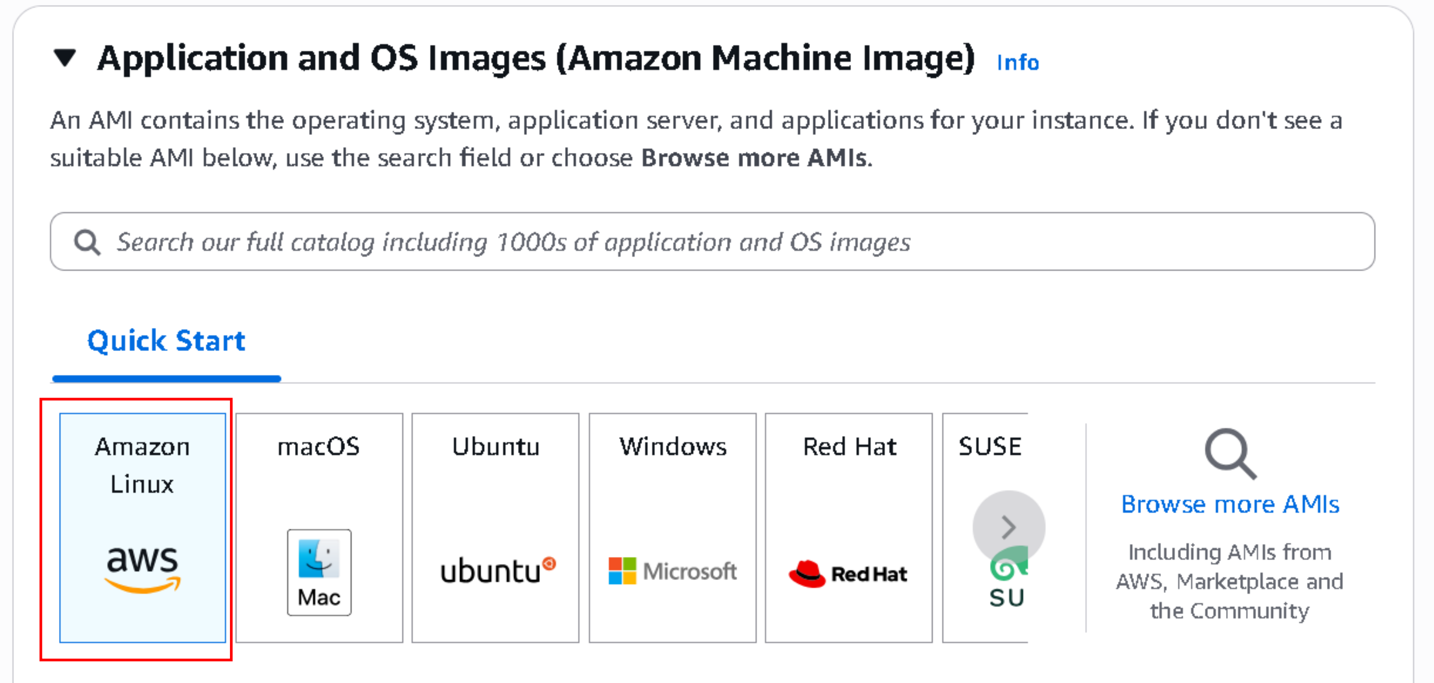
- Now simply scroll down and click Create Security group.
- Now our two Security group is ready,
- Now I am going to create two EC2 instance.



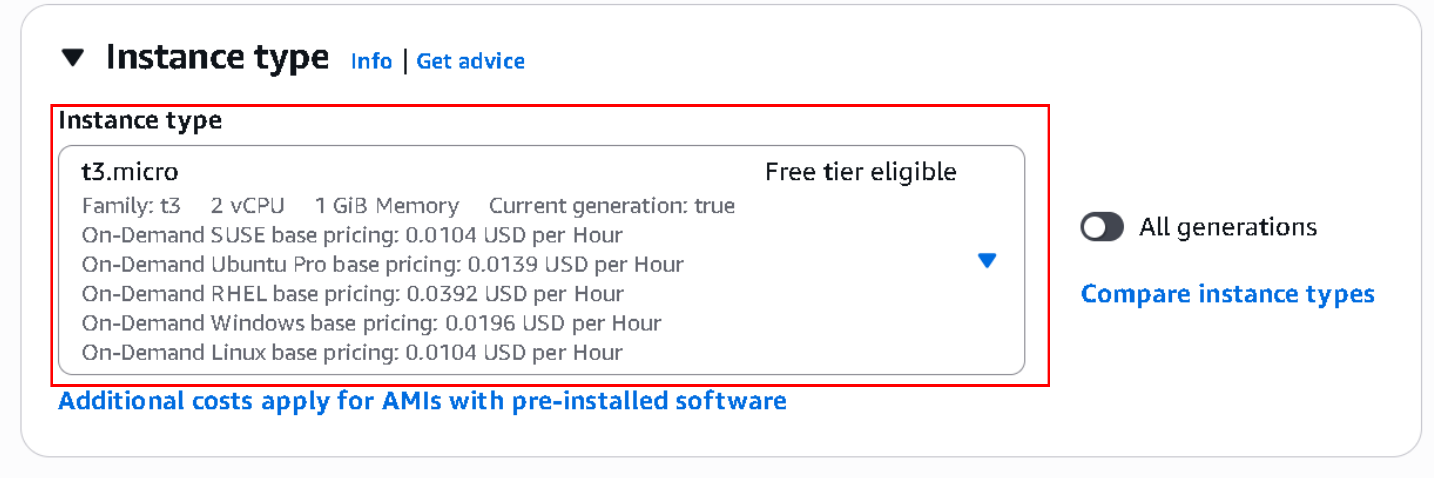
- Now opens EC2 in new tab.
- And click Launch instance.



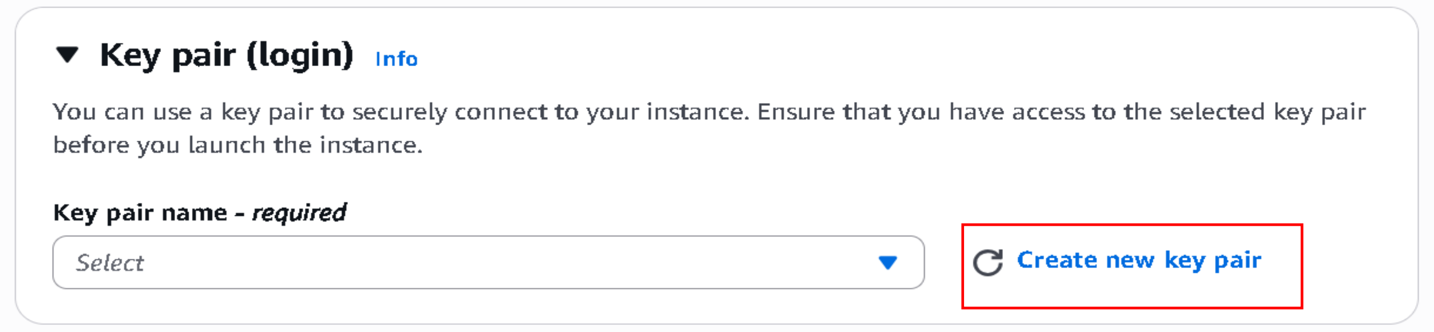
- Enter instance name (web-server-1).
- Scroll down.



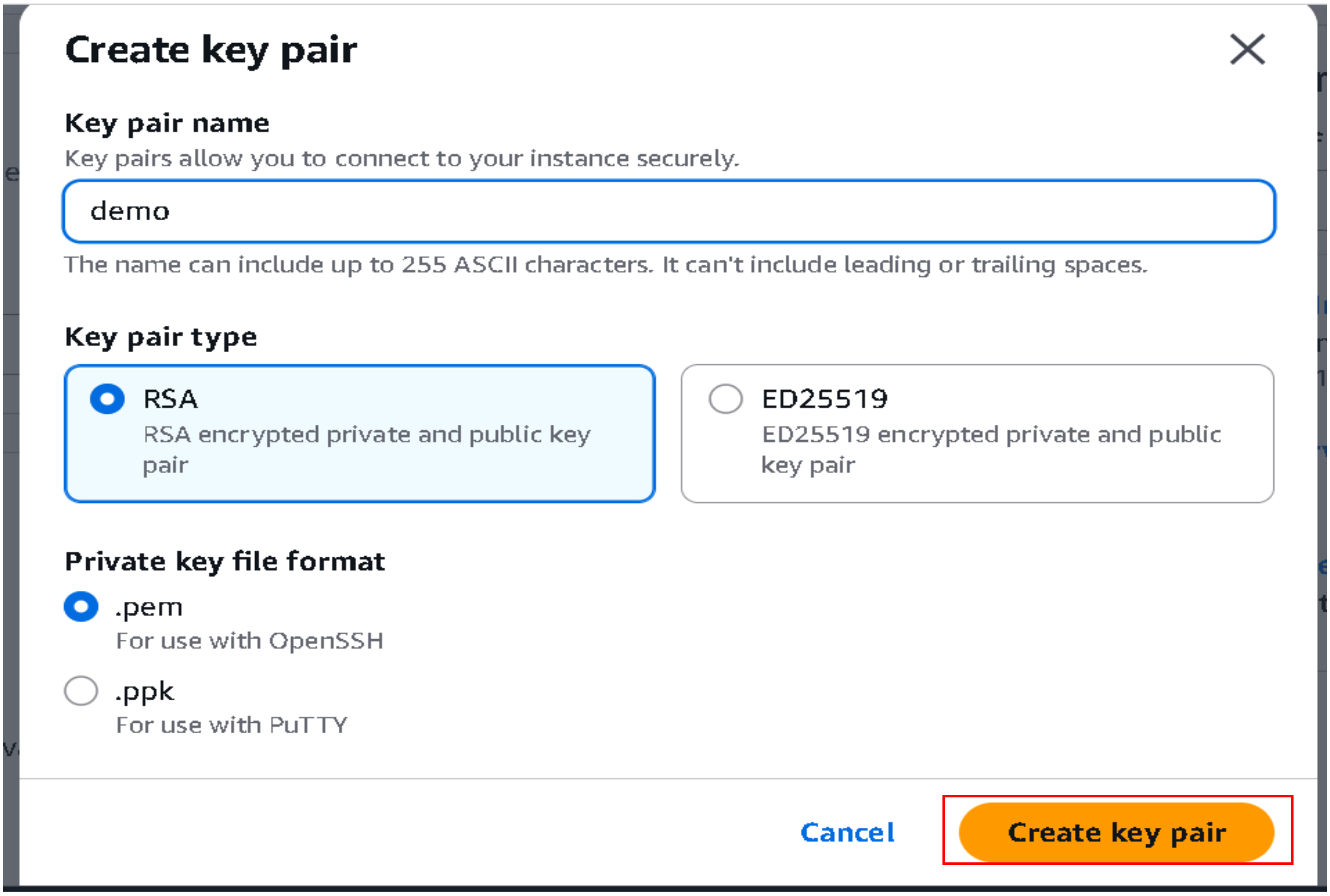
- Choose OS in my case I choose Amazon Linux.
- Scroll down



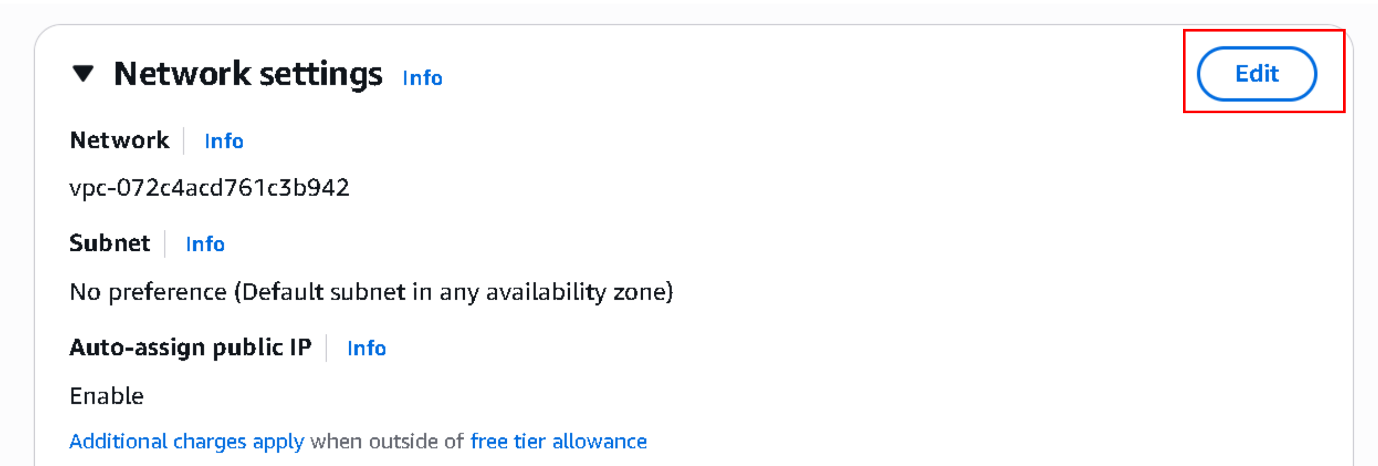
- Choose t3.micro this is Free tier eligible.
- Scroll down.



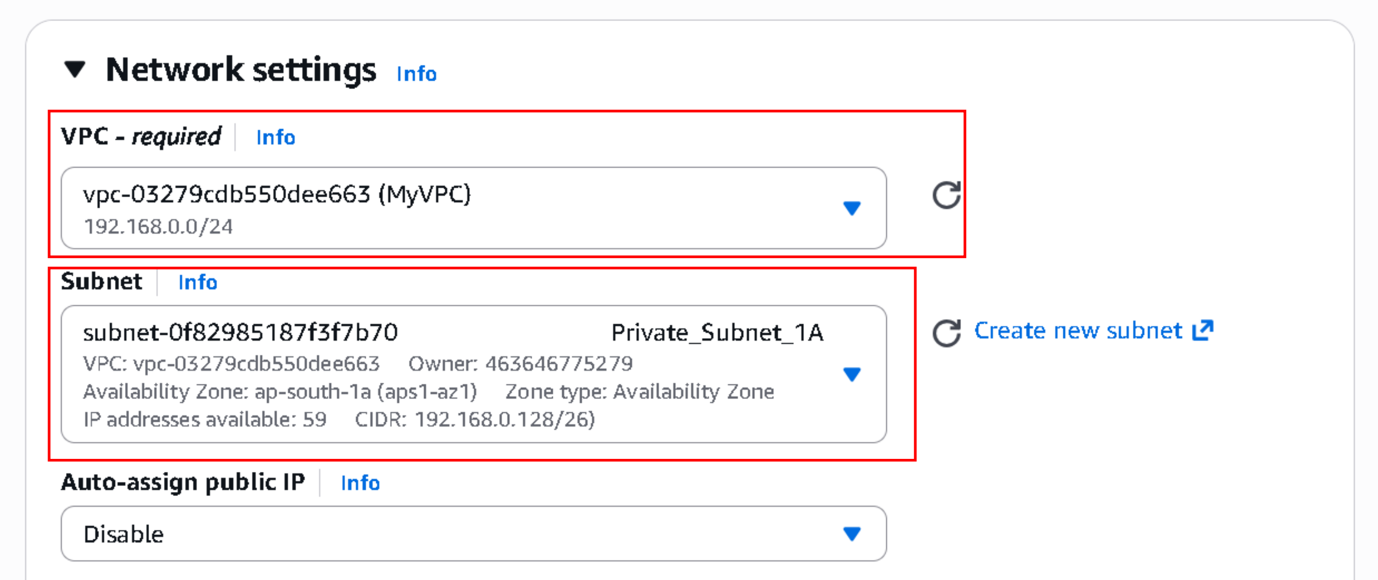
- Click Create new key pair.



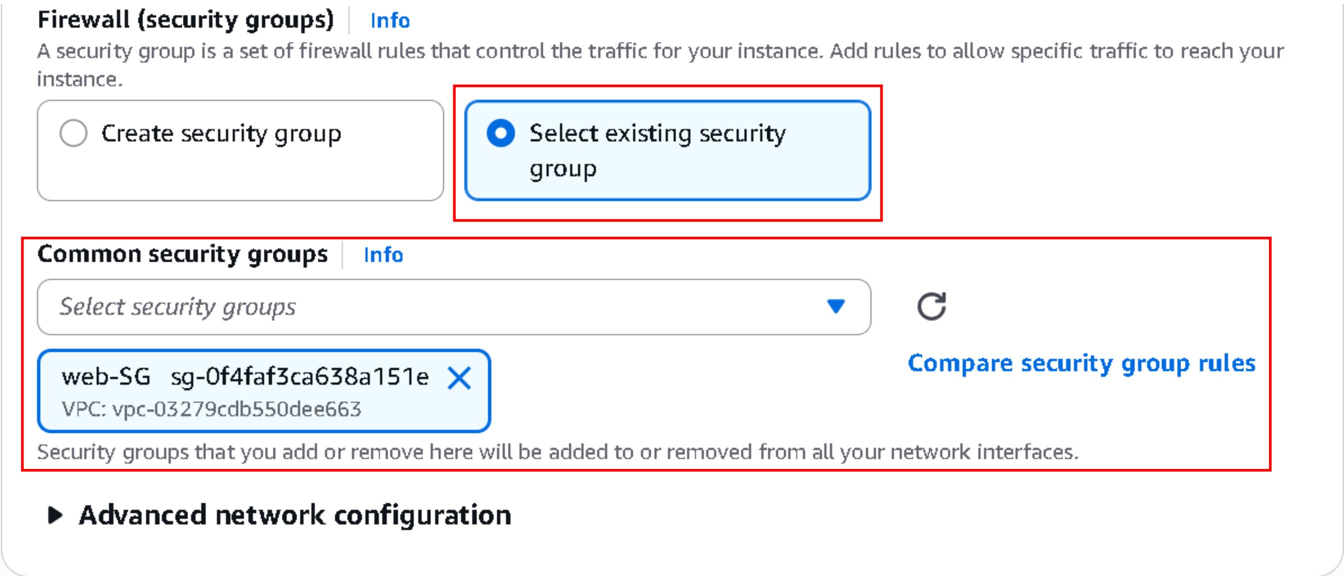
- Enter key name.
- Key pair type as RSA
- And .pem file
- And last click Create key pair
- Now your keypair is downloaded and save it.
- Now scroll down.



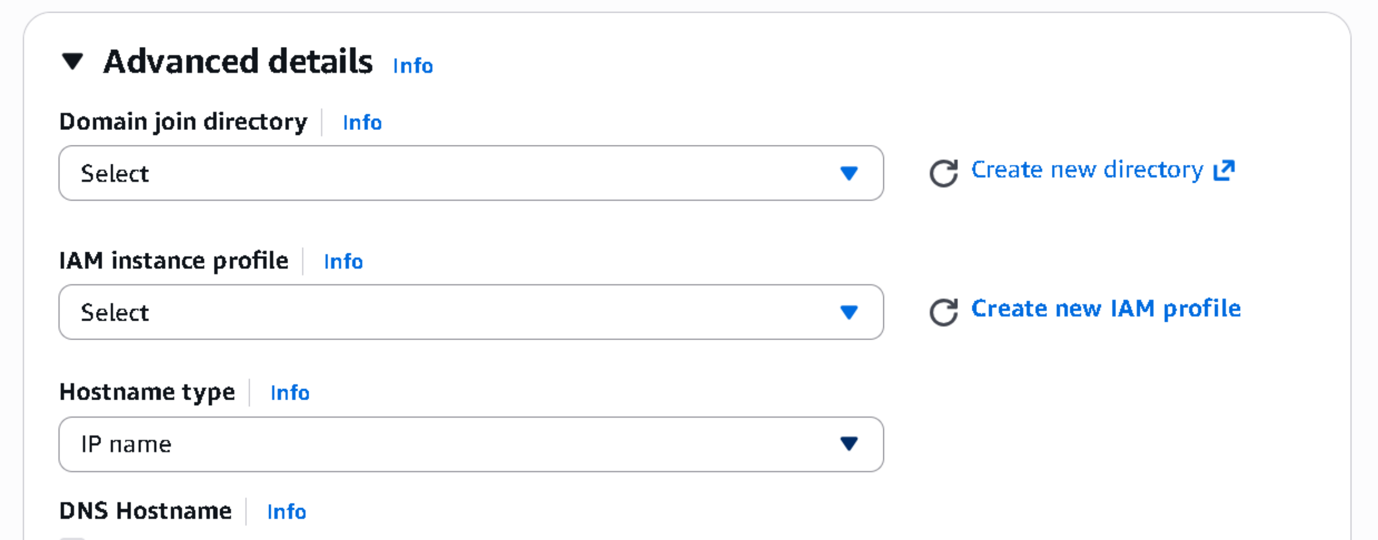
- Now simple Edit.



- Now select created VPC.
- And then Select subnet private_Subnet_1A.
- And Auto-assign public IP Disable.
- Scroll down.

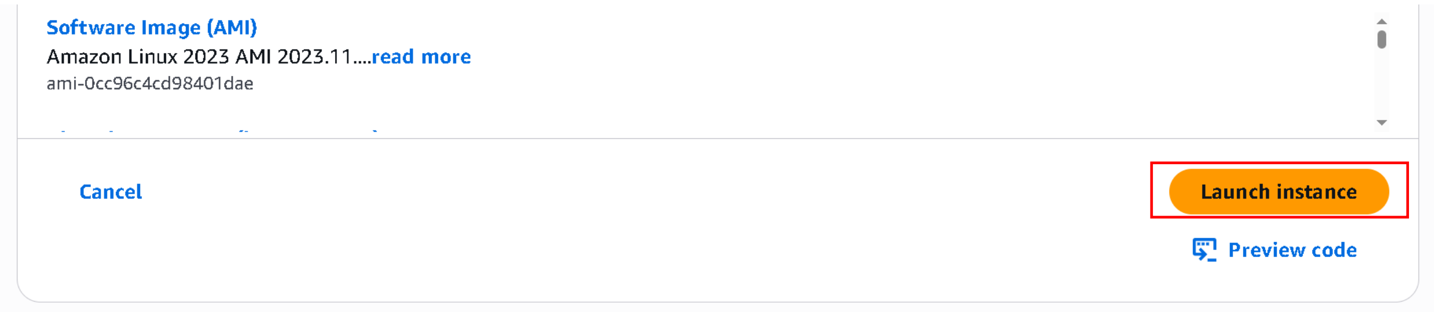


- Select Firewall (security group) existing security group.
- And select your created web-SG security group.
- Now we have our EC2 instance in a private subnet okay.
- Now how we will place our website over there because if you want to access this we have to create a EC2 instance connect End point in order to access this.
- Because our EC2 instance is in a private subnet and we don't have public IP.



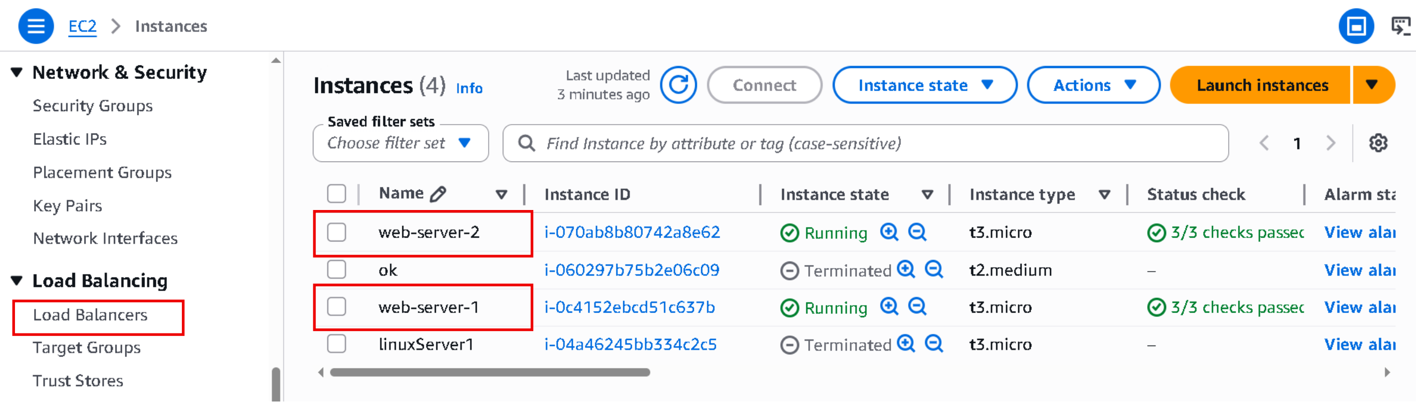
- So, for that scroll down and open Advanced details.
- And scroll down.

- Now here paste the script that provided in your resource.
- Now simply scroll down.

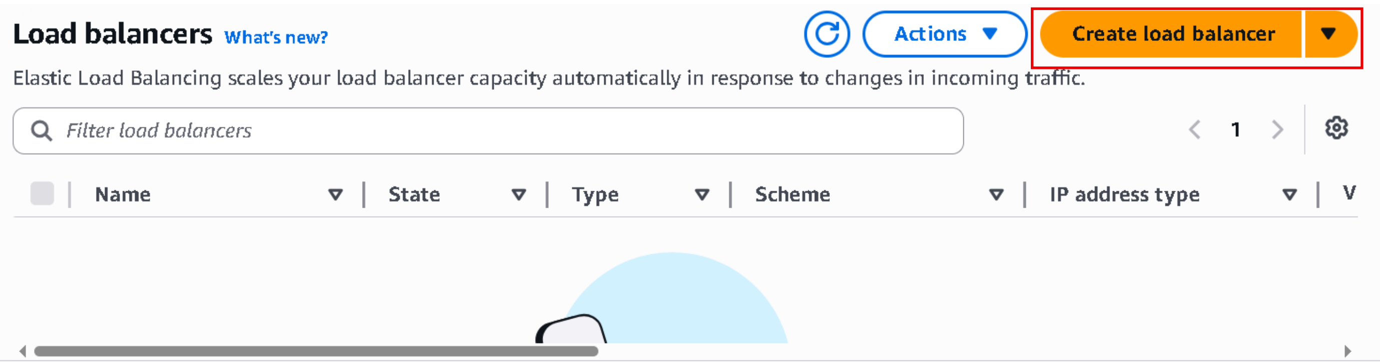


- Now leave remaining setting as it is.

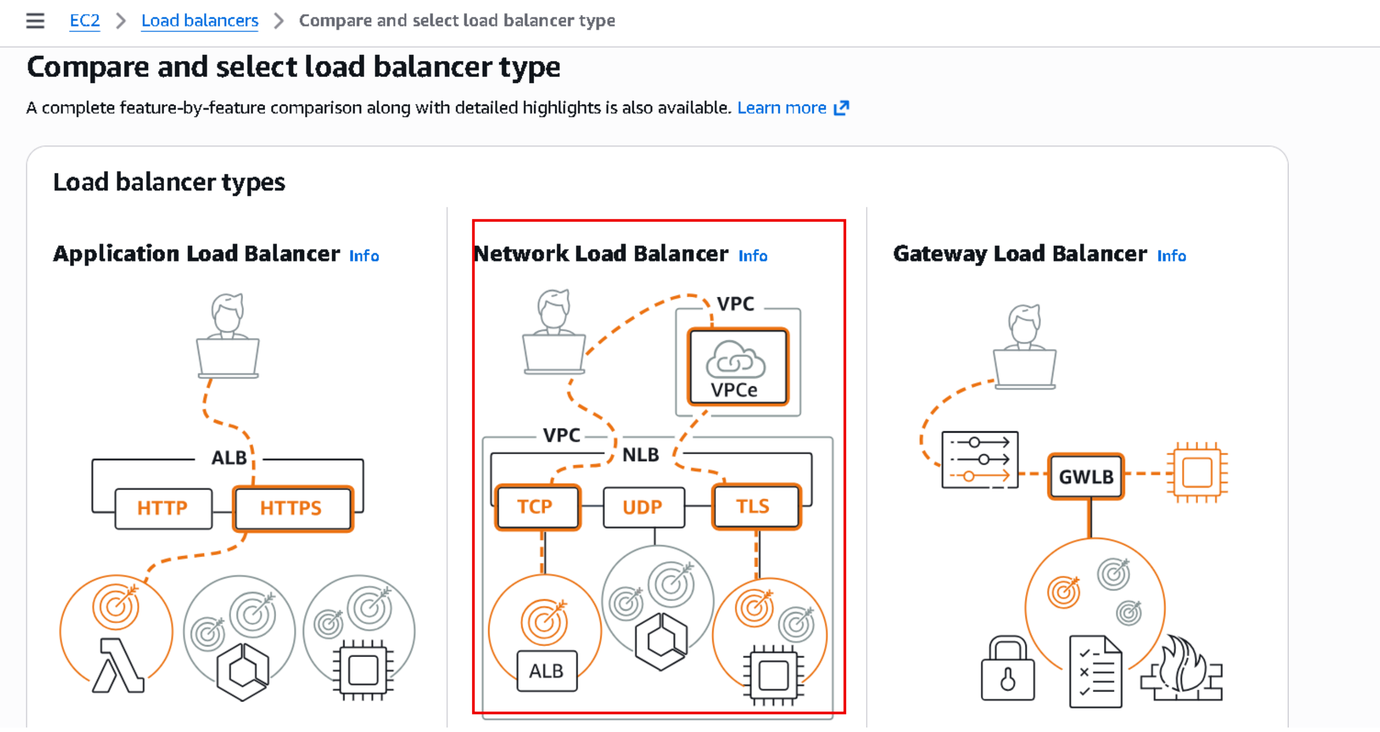
- And click Launch instance.
- Now we need to create 2nd EC2 name web-server-2 and same network setting only one change subnet is Private_subnet_2B
- And User data for 2nd EC2 instance is provided in Resource.



- Our both EC2 instance is created.
- Now click Load balancer for creating Network Load Balancer.

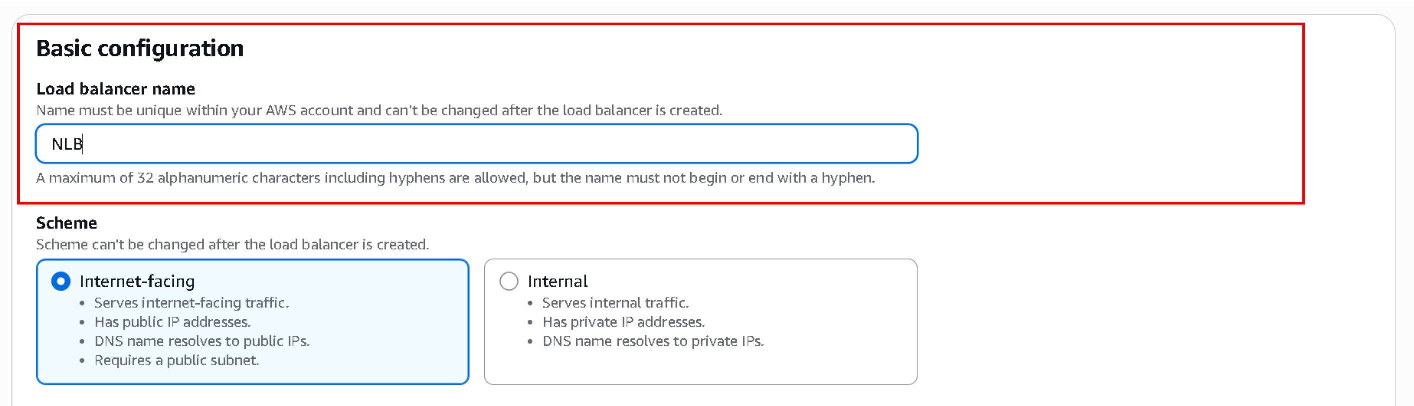


- Now simply click Create load balancer.

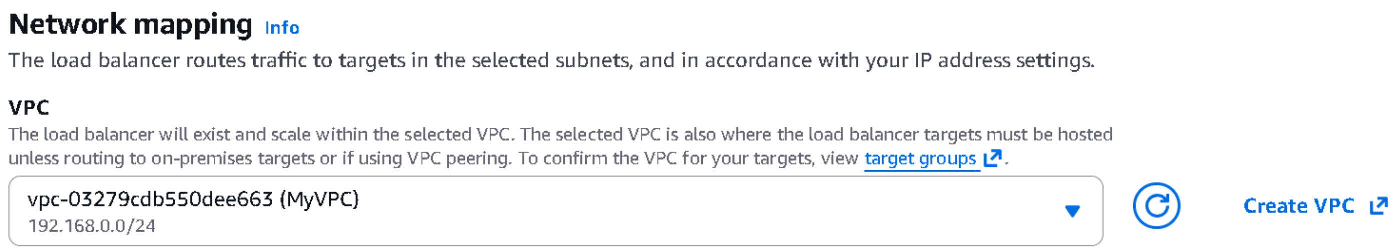


- Now we are going to create Network Load Balancer.

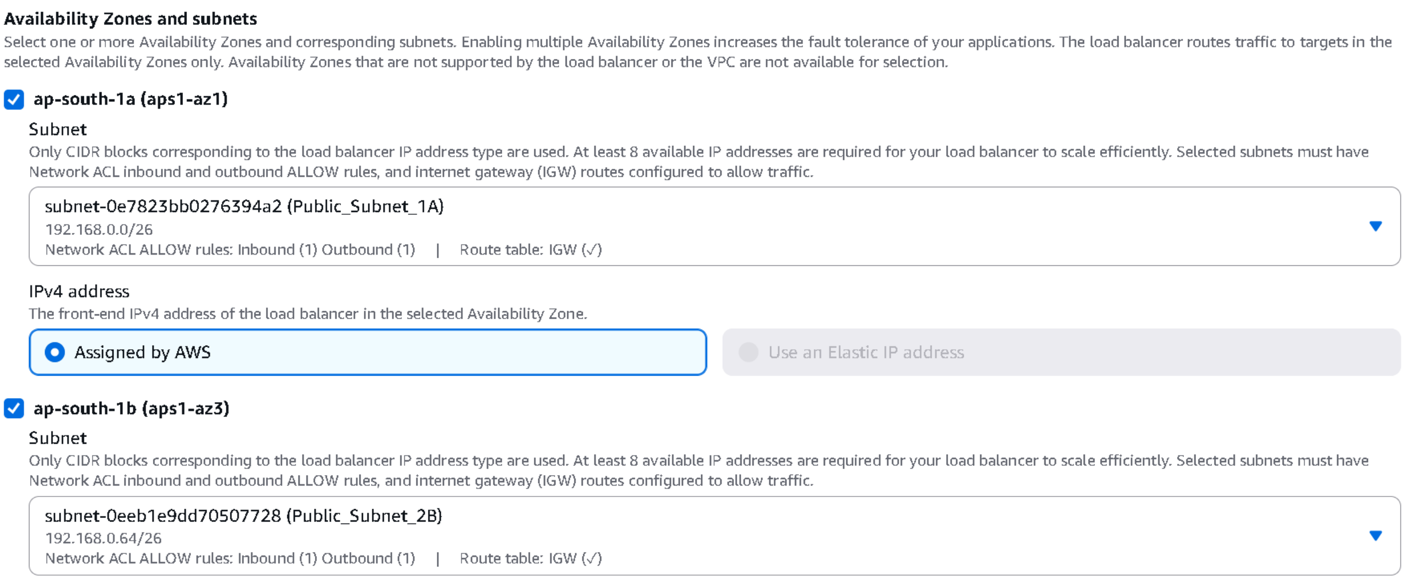
- Scroll down and click create.



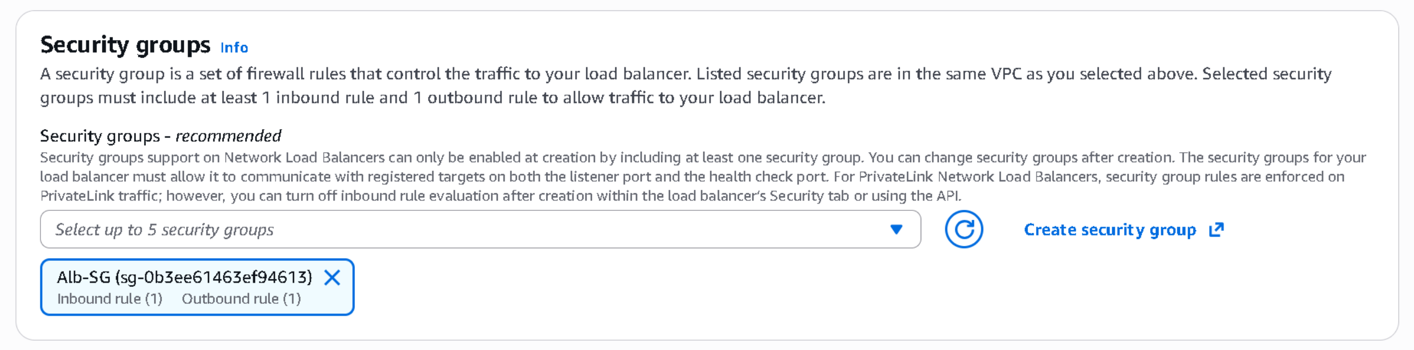
- Now Enter load balancer name (NLB).
- Scheme Internet-facing.
- Now scroll down.



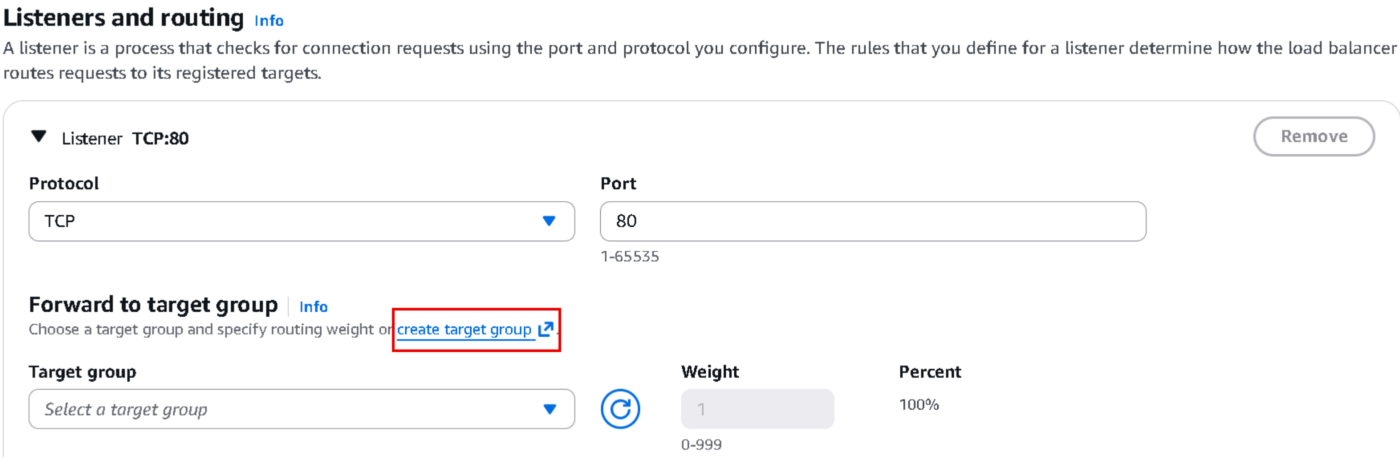
- And in Network mapping select your create VPC.
- Now scroll down.



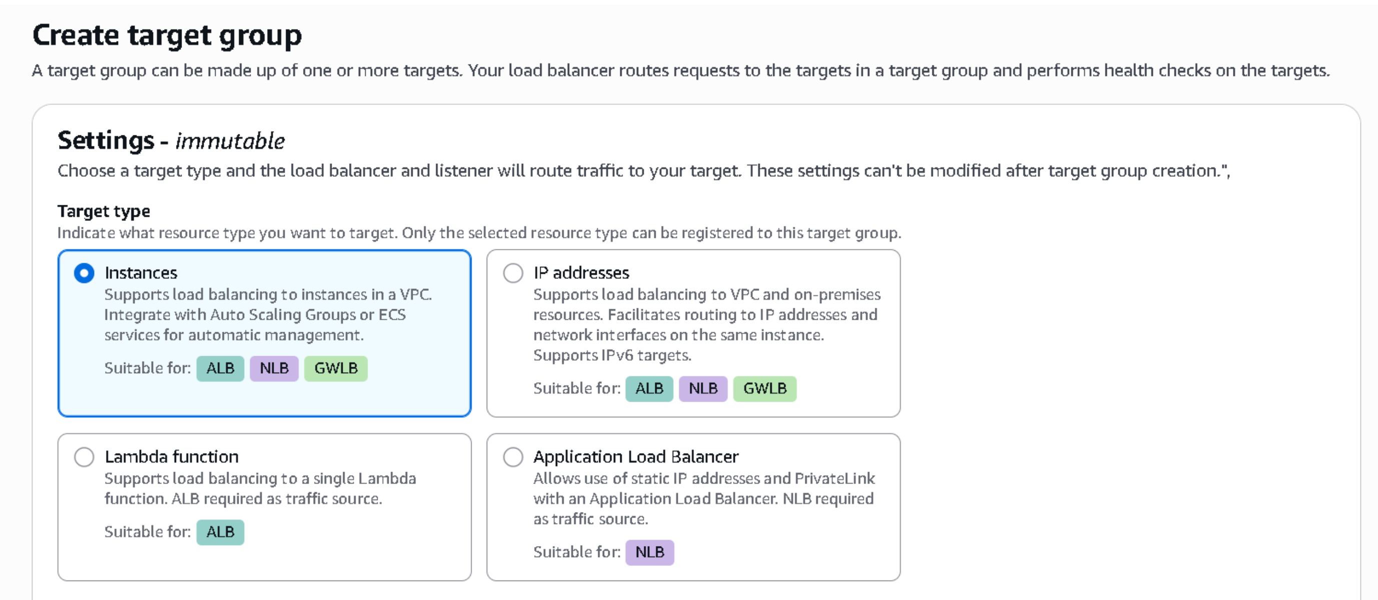
- Now select Availability zone and subnets.
- Select both Public_Subnet_1A and Public_Subnet_2B.
- IPv4 adddress Assigned by AWS (means ip address is dynamic).



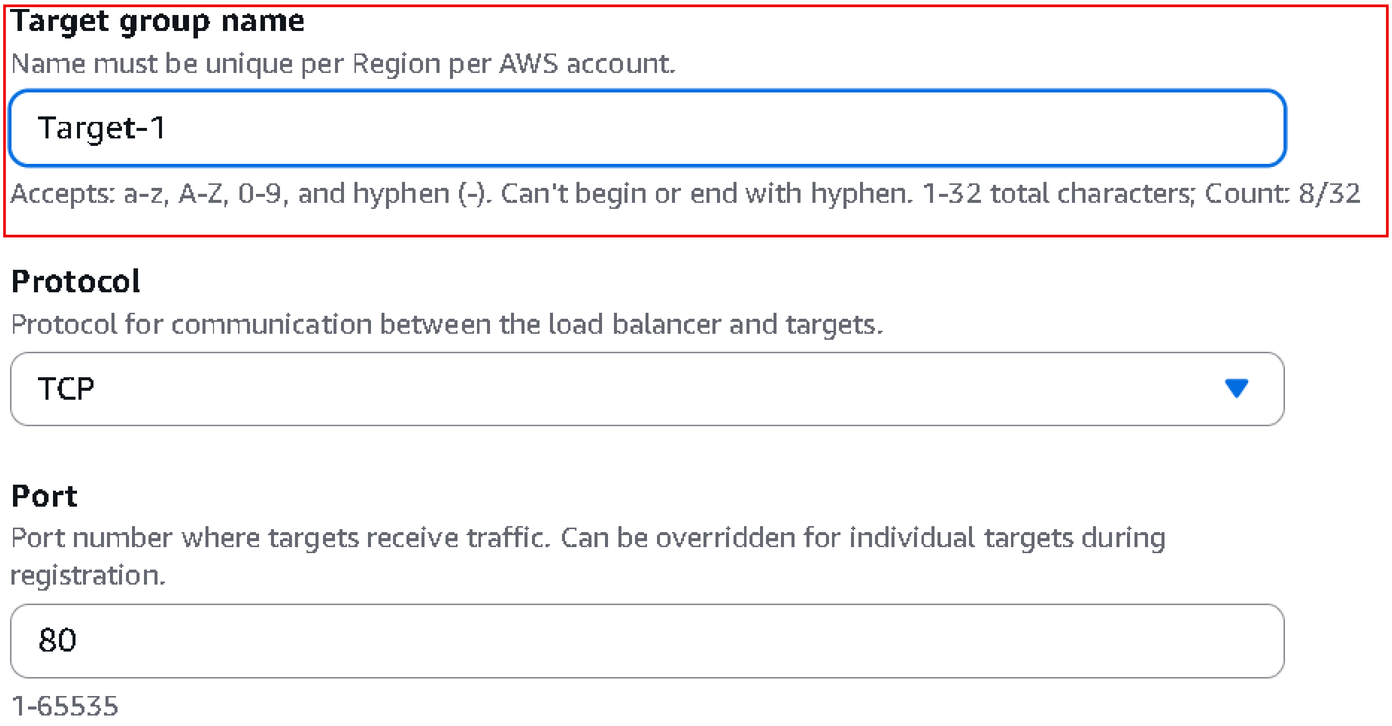
- Now select Created Security group (Alb-SG).



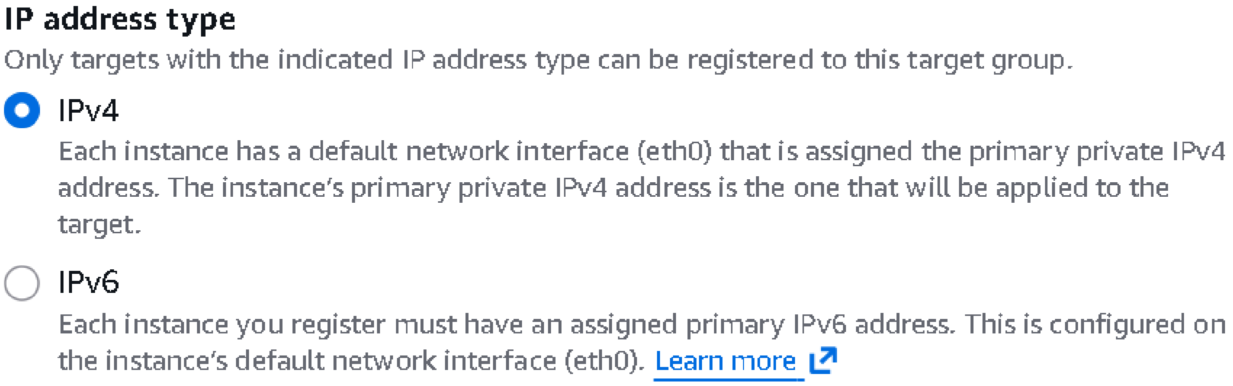
- Now protocol TCP and port 80.
- Now we need a target group.
- So, click Create target group.

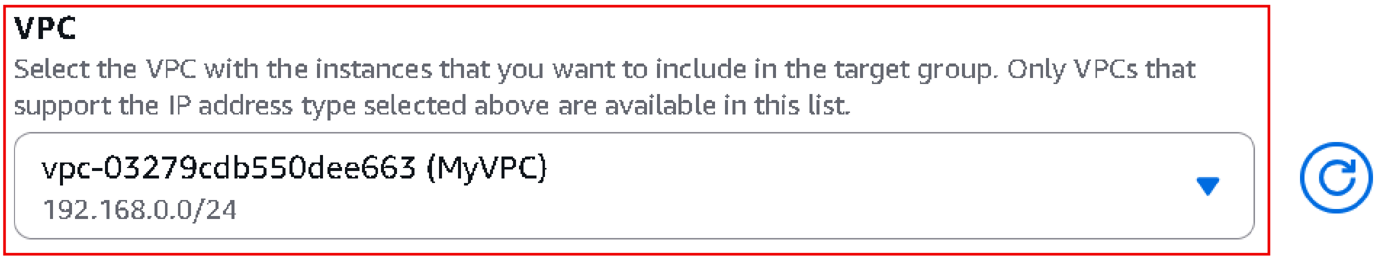


- Now select setting Instances.

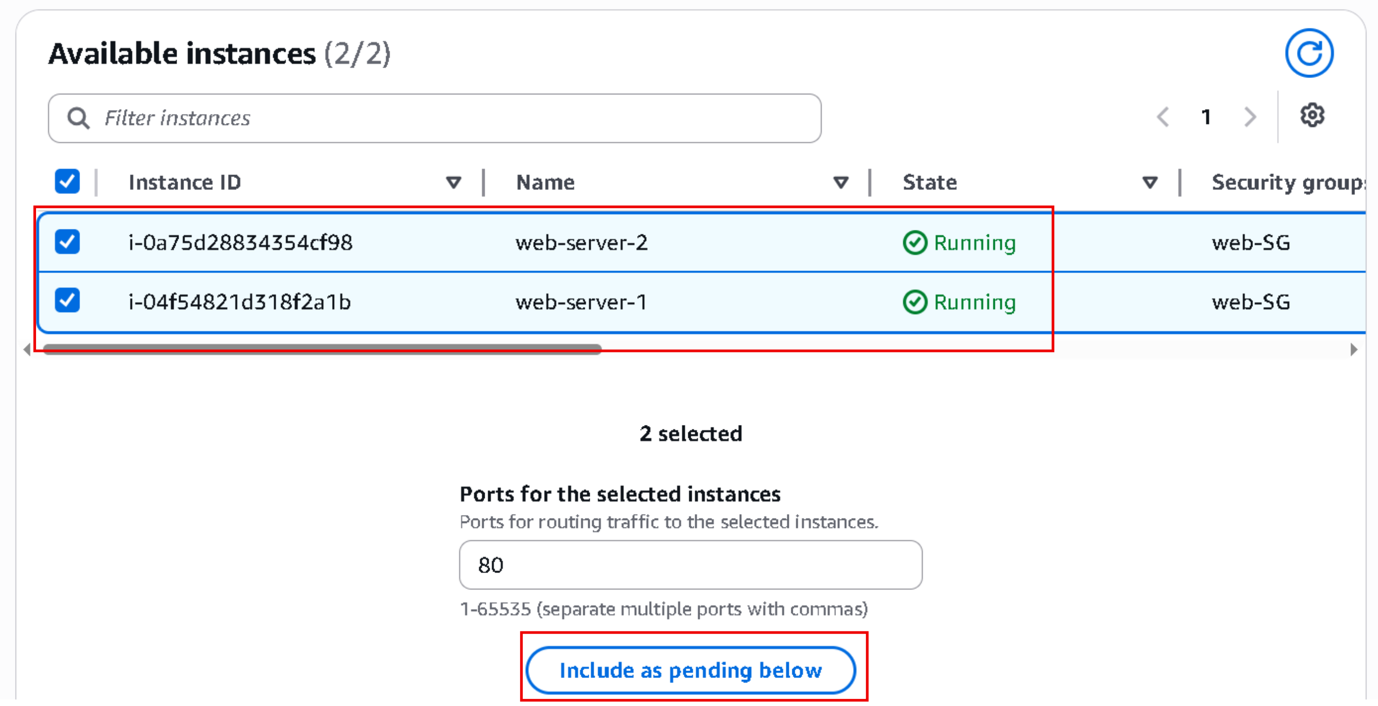


- Now Enter Name of target group.
- Scroll down.

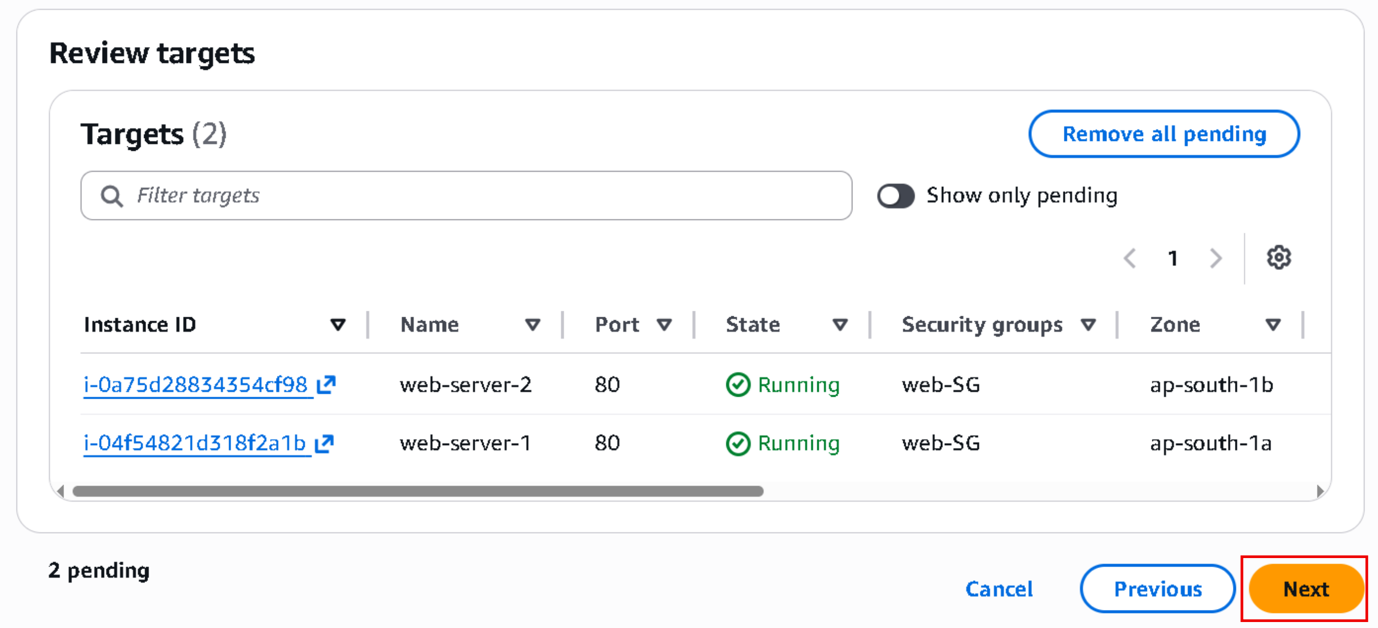




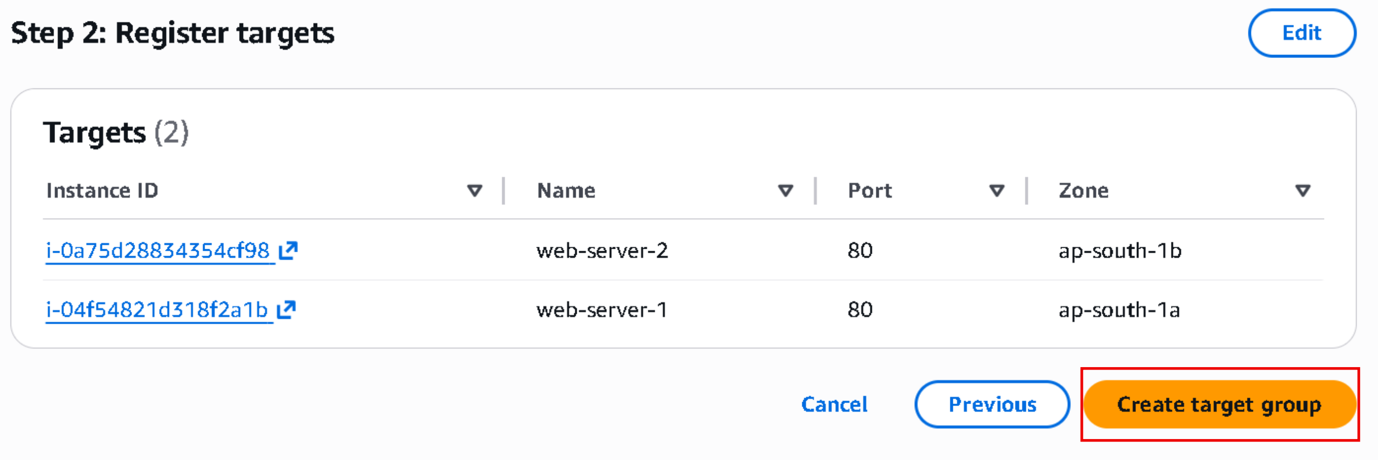
- Now Select the create VPC.
- And simply scroll down.
- And click Next.



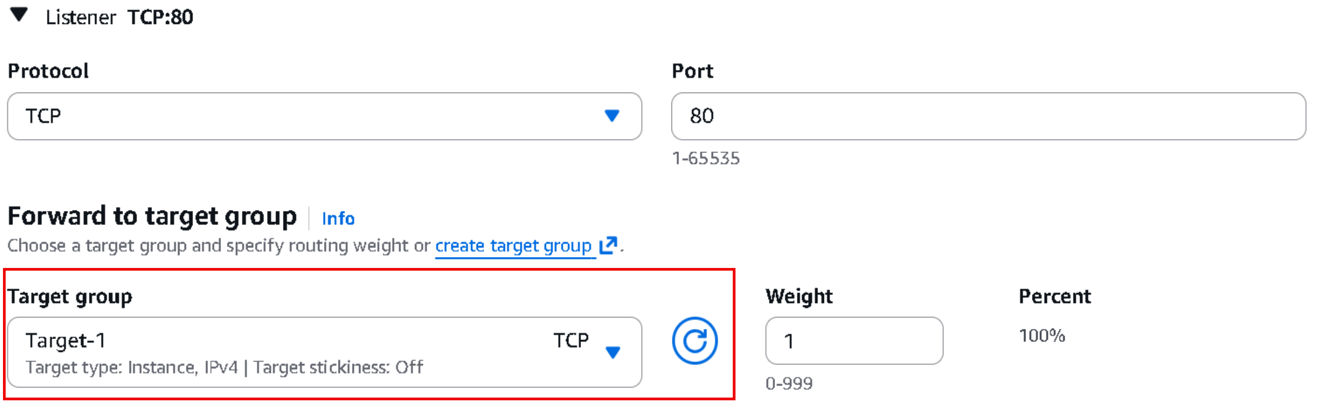
- Now select the Available instances that you created.
- And then click include as pending below.



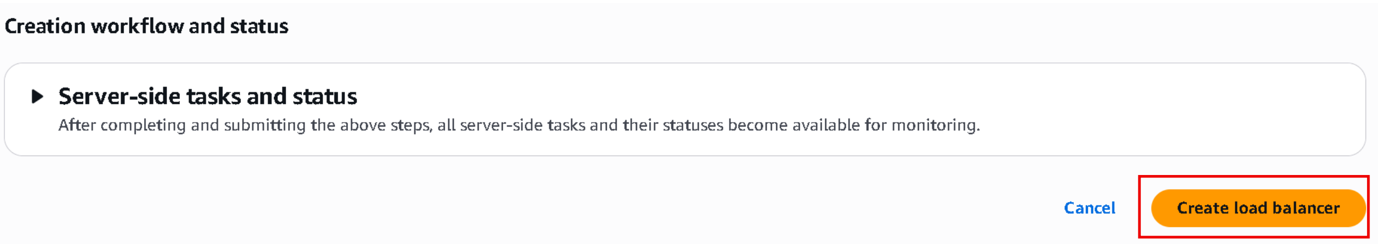
- Now simply click Next.



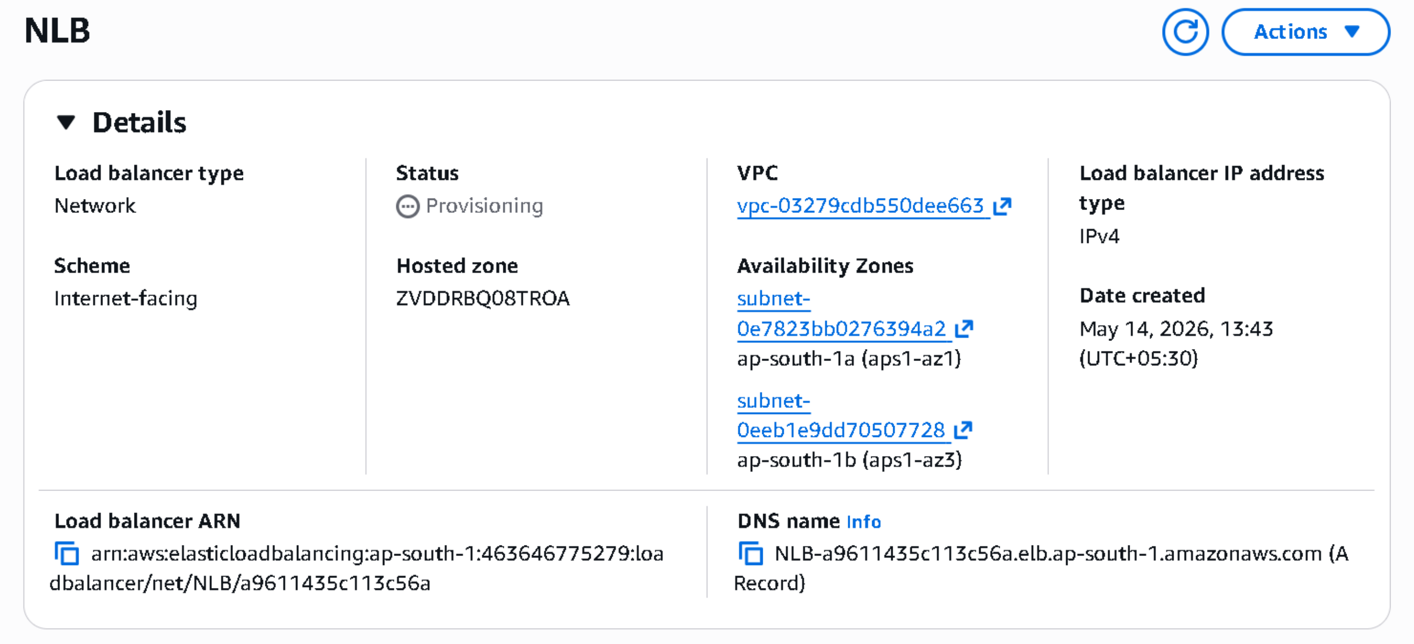
- Now simply click Create target group.
- Now our load balancer is created.
- And go back to Load balancer creation tab.



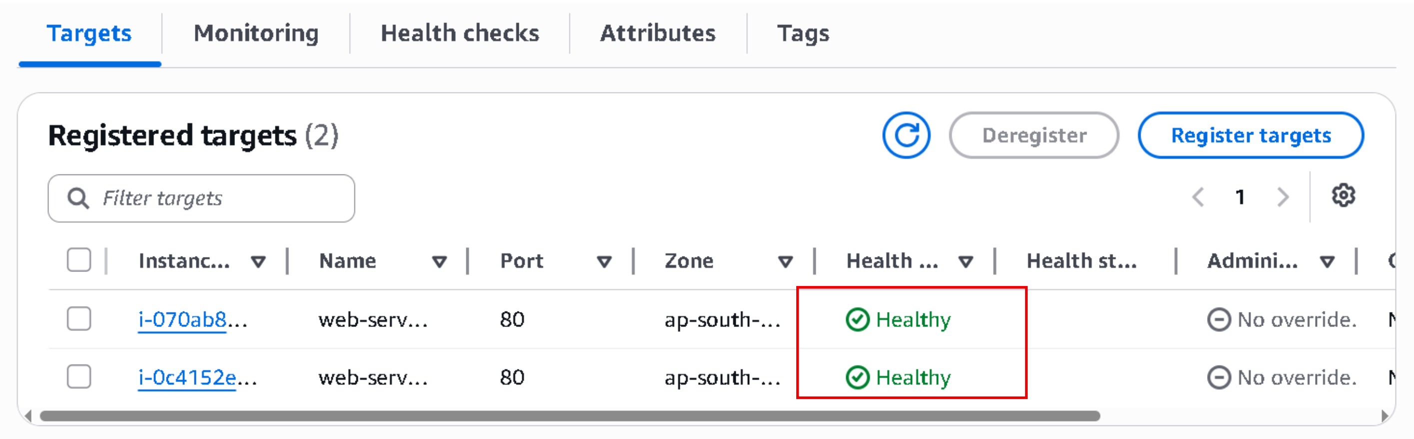
- Now click Refresh and select Target-1.
- Now simply scroll down.



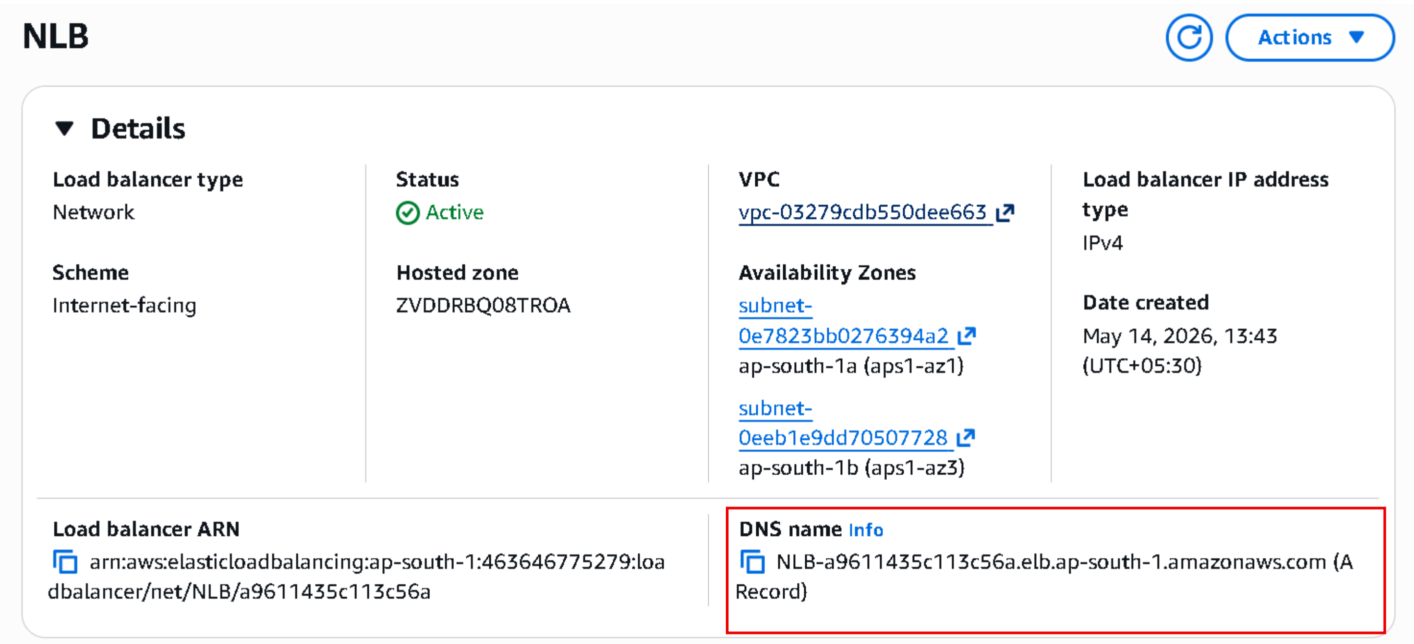
- And click Create load balancer.



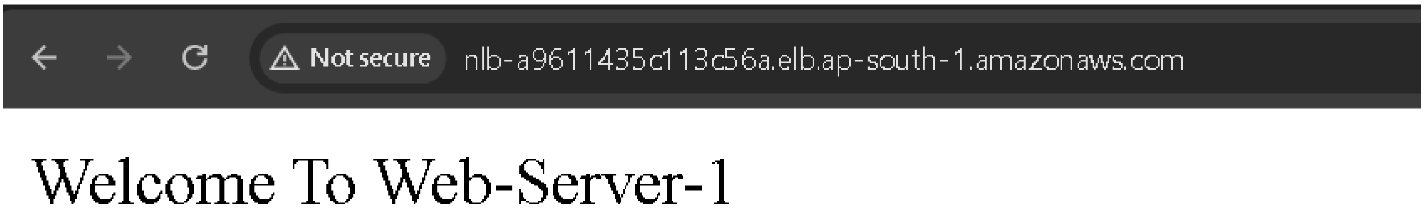
- Now NLB is Created you need to wait for its status become active.
- Then go to your target group and open created Target-1.



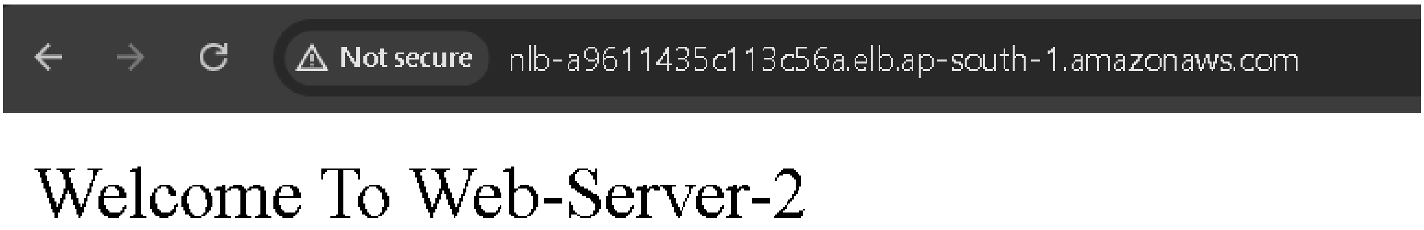
- Now simply scroll down and check health status.



- Now our Load Balancer status is active now copy the DNS name.
- And open it in new tab.



- Now you see its working.
- And when you refresh it shift dynamically load on both instance.
- Now refresh this tab.

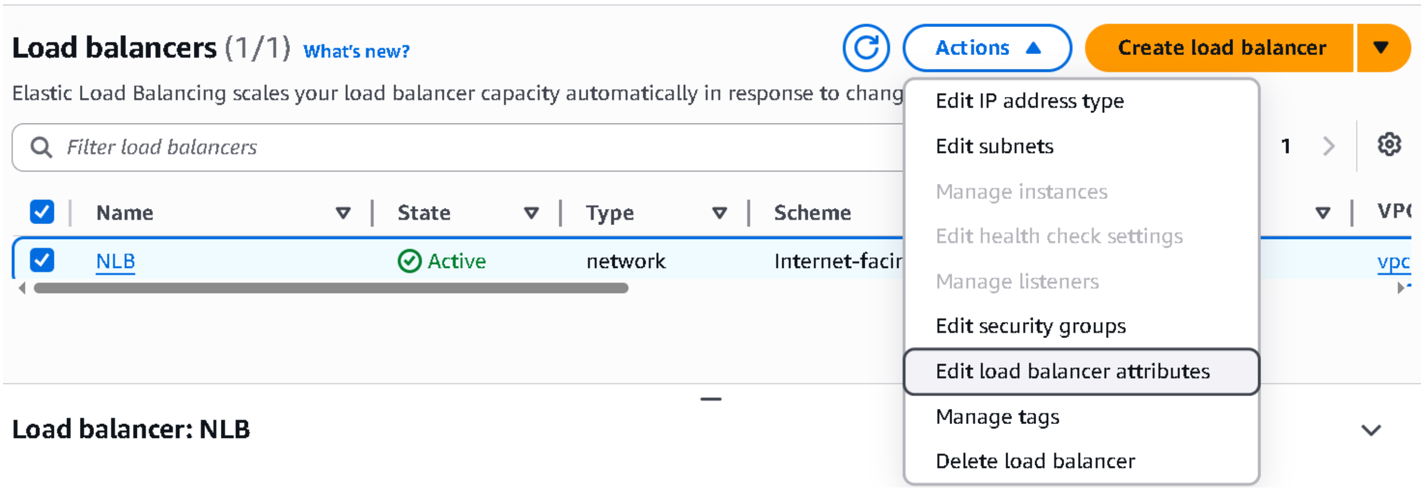


- Now you see it's balancing load on both instances.

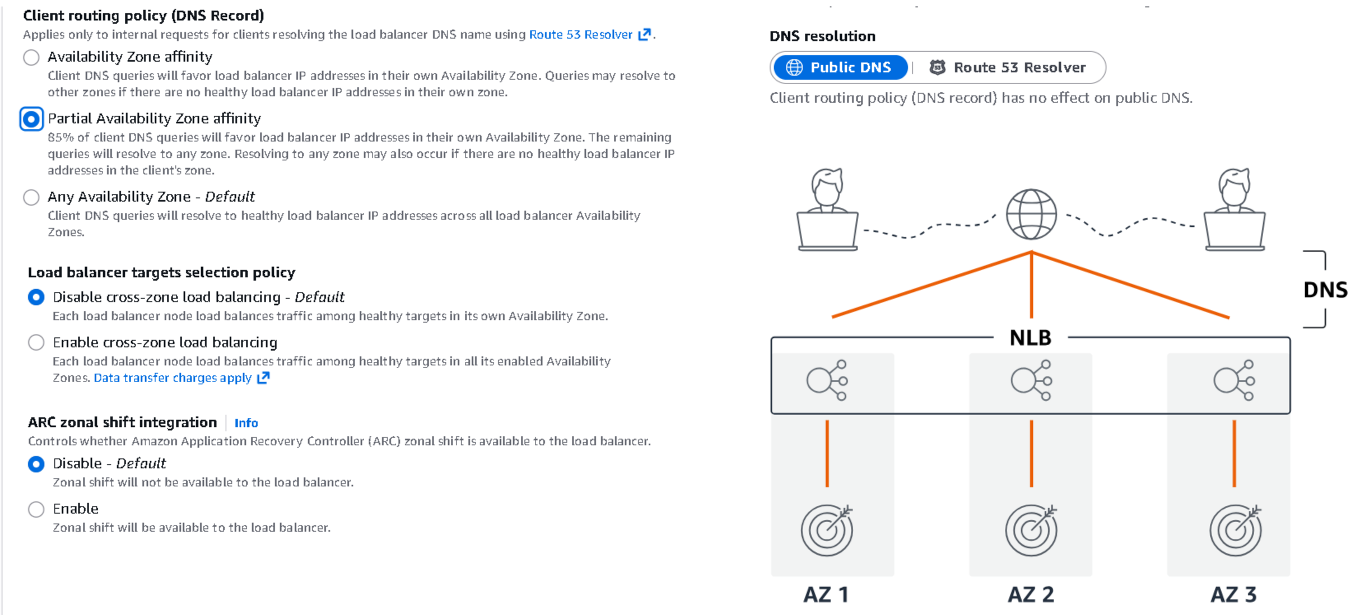
NOTE: - In Network load Balancer is working with layer four. Because it's working with layer 4 it is having ultra-low latency and it can handle traffic from millions of users, it prefers for gamming servers etc.

<u>Now let's implement cross Zone Load Balancing: -</u>

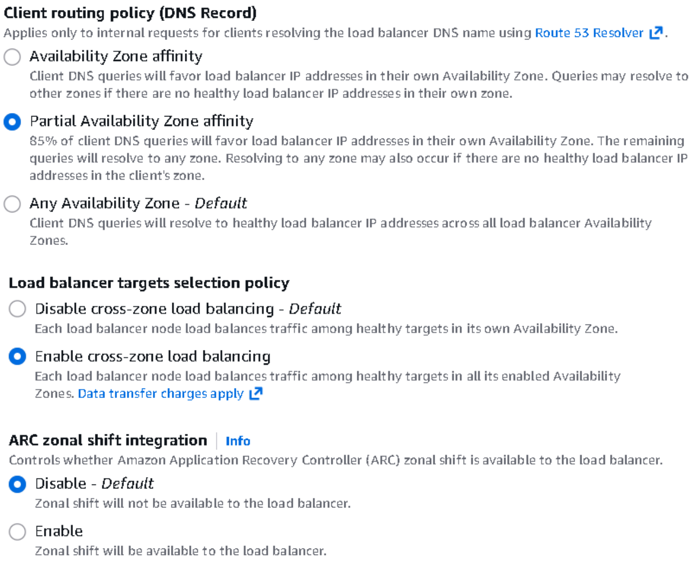
- It is the most important attribute of load balancer.
- Now open load balancer.

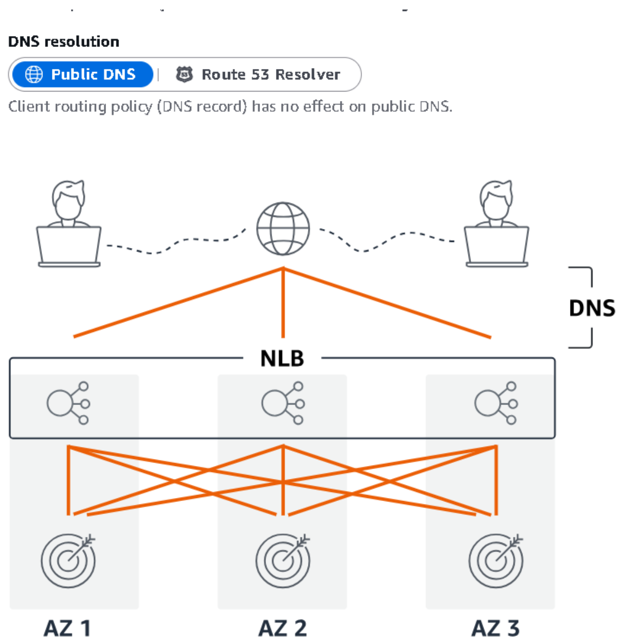


- Now select Created Load Balancer.
- And click on Action and select Edit load balancer attributes.



- Now, understand when you create a load balancer and select availability Zone in our case we select 2 availability zone south-1a and south-2b in creating load balancer.
- Means in each availability zone one load balancer instance is created.
- Now when you send request to your load balancer this request will be divided into two different node into the two different availability zone when cross -zone is disanbled.





- When we enable cross-zone load balancing.
- It means any load balancer in node in any availability zone will forward this request to any of the targets.
- It totally depends on you want to enable or disable.